

Joint scheduling of driver activities and charging for long distance electric truck operation under travel time uncertainty

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Abstract

Battery electric trucks are entering long-haul freight operations, but various additional constraints compared to conventional diesel trucks discourage freight companies from transitioning their fleets. As charging infrastructure expands, long-haul routes become increasingly feasible, and efficient schedules seek to synchronize charging stops with mandatory driver breaks. However, this synchronization remains fragile due to battery limitations and operational uncertainty, and becoming stranded due to lack of battery is unacceptable. This paper studies the joint scheduling of driver activities, such as breaks and rests, and partial non-linear fast charging activities for a battery electric truck serving a fixed sequence of customers on a long-haul route under travel-time uncertainty.

We formulate a mixed-integer linear program capturing the Hours of Service regulations, including split breaks, reduced rests, extended driving allowances, and the interaction between charging and break. Around this model, we build and compare six decision architectures that differ in when decisions are committed and what information they use: a scenario-based rolling-horizon look-ahead policy, a myopic greedy policy, a robust plan, a budgeted robust plan, a two-stage stochastic plan with online duration recourse, and a perfect-hindsight oracle. ~~Experiments on synthetic corridors and a real Norwegian long-haul instance show that~~ TO BE DONE

Experiments on 1 800 synthetic corridor instances show that adaptivity, not information, is what pays: the two policies that re-decide en route recover most of the hindsight optimum (5–13% mean gap) whereas plans that commit every duration before departure pay 23–35%, and the entire penalty of committing early is concentrated in customer time-window compliance (41% window hit rate for the box robust plan against 85% for the two-stage plan) rather than in driving time. Electrification itself costs 8–9% of makespan at the hindsight optimum: mandatory breaks do supply usable charging windows — 70–76% of charging time is credited inside a break — but fast-charging station access overheads absorb roughly half of that credit. An operator scheduling myopically observes 20–23% instead, so about half of the observed electrification penalty is a planning failure rather than a physical cost. On infrastructure, charger power dominates charger density by a factor of four at the spacing

mandated by Regulation (EU) 2023/1804.

► **COMMENT**

The abstract above is a *draft assembled from the numbers currently in the repository*. Every figure in it is traceable to `data_output/`, but all of them will move as the remaining runs land. Re-generate before submission. Note that the claim “and a real Norwegian long-haul instance” has been *removed*: that instance does not exist yet (Section 7.3).

Keywords: Electric trucks, Truck driver scheduling, Travel time uncertainty, Robust optimization, Rolling horizon

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Editorial annotation layer — read this first

This build of the paper carries a review layer on top of the author text. Nothing of the original wording has been silently changed: every intervention is colour-coded.

Colour	Meaning
Green	<i>Comment.</i> An observation, a justification, or a warning. Nothing to write, but something to know.
Red	<i>Rewriting / correction.</i> The existing text is wrong, stale or internally inconsistent. Red text is the proposed replacement; deleted text is struck out in grey.
Blue	<i>To be done.</i> A run, a check, a computation or a decision that is still outstanding.
Purple	<i>Suggested new writing.</i> Text that does not exist yet and that the paper needs. Drop-in prose, to be edited.
Orange	<i>Code vs. paper mismatch.</i> The manuscript describes something the implementation does not do, or vice versa. These are the most dangerous items: they are all falsifiable by a referee with the repository.

Where the numbers in this build come from

► COMMENT

The manuscript you supplied is *behind the repository*. Three generations of numbers are currently in circulation: (i) the numbers hard-typed in the manuscript, (ii) the numbers in `tex/sections/results_section.tex` (rewritten 2026-08-03), and (iii) the numbers in the generated files `tex/tables/*.tex` and `data_output/*.csv`, which are newer than both and are currently uncommitted. **This build uses (iii) throughout**, and marks in red every place where the manuscript prose disagrees with it.

Priority list

TIER 0 — blocking. A referee who opens the repository will find these.

1. Prose and Table 9 disagree on *every* headline reliability number (Section 8.2). Fix by emitting L^AT_EX macros from `compile_solutions.py` and never hand-typing a computed number again.
2. Table 8 reports cells computed from as few as 1 instance as if they were 50-realization class means. Now flagged in the table.
3. Survivorship bias is acknowledged but never quantified: 2SP is scored on the runs it survived, RO on all of them.
4. Constraint (was 54) (the 60 h weekly working cap) is in the paper but **explicitly out of scope in the code**.

5. Section 2.4 states the travel-time/energy coupling *backwards*. Under the paper’s own ECR curve a delayed leg consumes *less* energy.
6. The introduction claims “where to wait” is a decision. Idle waiting is disabled in the model.

TIER 1 — code/paper mismatches (all marked **orange** in situ): Γ , look-ahead action pruning, the partial LP relaxation, the queue model, charging-station spacing, customer counts, cluster geometry, the ECR reference, and $\mathcal{T}_{\min}$.

TIER 2 — the experimental design section does not describe the experiment that was run. Table 6 presents three bundled classes; the code runs a $3 \times 3 \times 4 \times 50 = 1\,800$ full factorial.

TIER 3 — missing work: contributions, conclusion, Norwegian case study (skeleton only, no data), VSS/EVPI (harness written, zero runs), captions for Figures 1–4, six broken citations, one broken cross-reference.

TIER 4 — prose contradicted by the regenerated tables: the long-route diesel oracle, 2SP on long routes, the refuelling assumption, and which methods appear in the sensitivity analysis.

TIER 5–6 — contribution framing, positioning table, statistics, and figure style.

1. Introduction

Road freight accounts for roughly a quarter of transport CO_2 emissions in the European Union, and battery electric heavy-duty trucks (BEHDTs) are the leading decarbonization pathway for the segment. Their deployment in long-haul service, however, is first constrained by battery capacity and charging infrastructure, but also by an operational factor that is absent for passenger electric vehicles: driver’s regulations. Commercial drivers in the EU operate under Regulation (EC) No 561/2006 and Directive 2002/15/EC, which impose mandatory breaks after at most 4.5 h of driving, daily rest periods, and limits on shift driving time. Because a mandatory 45-minute break is long enough for a substantial fast charge, synchronization of charging and drivers constraints is critical. Whether electric long-haul operation can match diesel route durations therefore hinges on how well charging events can be synchronized with legally required interruptions.

However, this synchronization is fragile. Travel times on long-haul corridors are uncertain, and a delay on a single leg propagates through the entire schedule: it consumes driving-time budget before the planned break location is reached, shifts arrivals outside customer service windows, and, because energy consumption depends on realized speed, perturbs the state of charge with which the truck reaches the next charging opportunity. A pre-determined schedule that nests charging inside breaks under average travel times may, in execution, force the driver into an unacceptable event such as running out of battery if the energy consumption deviates from anticipated values.

► CORRECTION / REWRITE

The last sentence is right but the mechanism is stated loosely, and in Section 2.4 the same mechanism is stated *backwards*. Replace the second half of this paragraph with a statement of the actual coupling, which is one of the more interesting features of the problem:

“...and, because energy consumption depends on realized speed, perturbs the state of charge with which the truck reaches the next charging opportunity. Crucially, the two risks pull in *opposite* directions: a slow leg consumes driving-time budget and threatens Hours-of-Service compliance and the customer window, whereas a *fast* leg is the one that threatens the state of charge, because the energy consumption rate rises with speed over the whole operating band of a heavy vehicle. A schedule that is protected against delay is therefore not automatically protected against depletion, and vice versa.”

► COMMENT

Why this matters beyond correctness: it is the justification for the two-sided budget in Section 6.5, and it is currently the only place in the paper where the ROBU construction is motivated by the physics rather than by convention. Quantitatively, on a 25 km leg with the calibrated ECR curve, $\xi = \underline{\xi}$ (90 km/h) gives 29.4 kWh, the nominal 80 km/h gives 27.5 kWh, and $\xi = \bar{\xi}$ (50 km/h) gives 26.7 kWh — the energy span across the *entire* support is +7% / –3% around nominal.

This paper studies the joint scheduling of driver activities and charging for a battery electric truck on a fixed long-haul route with multiple customer stops, under stochastic travel times. Given the sequence of customers, the decisions are where and for how long to

charge (with a nonlinear, piecewise-linearized charging curve and partial charging), where to take breaks and daily rests (including split breaks, reduced rests, and extended driving allowances, together with the rule that a sufficiently long charging stop can count as a break), so as to minimize route duration while respecting the daily provisions of EU HoS regulation, battery limits, and customer service windows.

► COMMENT

Already fixed in your source. The compiled PDF still listed “and where to wait” among the decisions; your `.tex` has removed it. Correct — `MILP.py` declares an idle-wait variable w_i but fixes it to zero unless `data["allow_wait"]` is set, which happens only in the R15-PGLT benchmark harness of Section 8.1.

What is still missing is the positive statement: “no idle waiting” must be added to the assumptions list, because it is load-bearing in Section 8.3.3, where the entire time-window mechanism is “the diesel arrives early, cannot wait, and therefore must stretch a break”. A reader who does not know waiting is prohibited cannot follow that argument.

Because decisions are taken sequentially while uncertainty is revealed leg by leg, the central methodological question is about the decision architecture: when should each decision be committed, and what information should it use? We organize our solution methods along exactly this axis. Three offline methods that commit a plan before departure: a box robust plan over an interval uncertainty set, a budgeted robust plan that protects only against a bounded number of simultaneous worst-case legs, and a two-stage stochastic plan whose activity structure is fixed but whose durations adapt online, and two online architectures that re-decide at every stop: a scenario-based rolling-horizon look-ahead policy and a myopic greedy policy representing typical driver behavior. All methods are evaluated in a common discrete-event simulation, and benchmarked against a perfect-hindsight oracle, so that the value of information and the value of adaptivity can be measured directly.

► TO DO

“so that the value of information and the value of adaptivity can be measured directly” is a promise the paper does not currently keep. The VSS/EVPI decomposition that would deliver it (Section 8.3.4) has a working harness (`src/output_analysis/vss_evpi.py`) and **zero completed runs**. Either run it, or downgrade this sentence.

The contributions are as follows. First, we are the first to model the full ~~weekly~~**daily** provisions of EU Hours-of-Service regulation **jointly with** partial charging on a non-linear charging curve **for an electric truck on a multi-customer long-haul route**, and **we** formulate a MILP **whose Hours-of-Service core is** validated against an independent exact method. Second, we analyse the impact of travel-time uncertainty through an extensive set of methods that **differ in** decision architecture, quantifying **what it costs to commit** to a schedule upfront **rather than to adapt** en route **as information is revealed**. Third, we quantify the electrification penalty of long-haul routes by decomposing it into charging time, **the additional station access it requires, and the break time that charging discharges**. This confirms the finding of Brockmann and Guajardo [6] that mandatory breaks create sufficient charging windows most of the time, with charge/break coupling reaching 80%. Finally, **we derive**

policy insights on charging-infrastructure investment for long-haul corridors: in particular, charger power dominates charger density by a factor of roughly four in route duration.

► CODE vs. PAPER MISMATCH

“weekly” must become “daily”. MILP.py lines 1241–1250 put weekly accounting explicitly out of scope: no weekly rest, no 56 h weekly driving cap, and constraint (was 54) is not implemented. This is the *first sentence of the contribution list*, so it is the first claim a referee will check, and Garaix et al. [14] already own the weekly TDSP. Daily-only remains a first in this combination, so nothing is lost.

► COMMENT

Two smaller points on the third contribution.

(i) “wait time” in the original draft has been replaced above: there is no wait time in the decomposition — idle waiting is fixed to zero and the row reads 0.00 in every class. The three real non-charging terms are queueing, manoeuvring and bay repositioning, and the term that makes the decomposition worth reporting is the *negative* one, the break the driver no longer takes separately.

(ii) “confirms” is your call, but it undersells the result. The decomposition shows the access overheads absorb roughly half the coupling credit, which is a *qualification* of Brockmann and Guajardo [6] rather than a confirmation — and a quantitative correction to the closest prior work is worth more to a referee than an agreement with it. A middle option that keeps the friendlier framing: “This confirms the finding of Brockmann and Guajardo [6] that mandatory breaks create sufficient charging windows most of the time, with charge/break coupling reaching 80%, while showing that the access overheads of a fast-charging stop, which that work does not price, absorb about half of the resulting credit.”

► TO DO

Three claims in the paragraph above still need a number or a check before submission. (i) “validated against an independent exact method” — say *39 of 40 published optima reproduced, and the 40th improved on*; the bare word “validated” wastes the strongest sentence available to you (Section 8.1). (ii) “coupling reaching 80%” — the class means are 68–76% and 80% is the maximum over time-window cells; make sure the text says which. (iii) “a factor of roughly four” — from Section 8.3.1, 4.3% against 1.2–1.5%, so the factor is three to four; “roughly four” is the optimistic end of that range.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 defines the problem and states the modeling assumptions. Section 4 presents the deterministic MILP. Section 6 describes the simulation framework and the six decision architectures. Section 7 details the experimental design, and Section 8 reports the computational results. Section 9 concludes.

2. Literature review

2.1. The truck driver scheduling problem

Hours-of-service (HoS) regulations impose constraints on the working and driving schedule of commercial vehicle drivers. In the European Union, Regulation (EC) No 561/2006

[11] specifies maximum consecutive and daily driving times, mandatory break structure and minimum daily rest periods. A similar directive (2002/15/EC) limits total shift working time and restricts work during night hours. Together these rules define the truck driver scheduling problem (TDSP): given a fixed sequence of customer stops, find a feasible and cost-efficient driver schedule complying with all regulatory constraints.

The first work to model government HoS constraints in a scheduling context is Xu et al. [42], who study a rich pickup-and-delivery problem under US rules and conjecture that finding a minimum-cost compliant schedule is NP-hard with multiple time windows. Archetti and Savelsbergh [2] show that the US TDSP with single time windows is solvable in $O(n^3)$ time. For European rules, Drexel and Prescott-Gagnon [8] formalize EU HoS as resource constraints in the elementary shortest path problem with resource constraints (ESPPRC), develop exact and heuristic labeling algorithms, and conjecture NP-completeness. Goel [16] proposes a depth-first label-setting procedure for EU rules, which is extended in Goel [17] to cover extended driving allowances and reduced rest exceptions. Kok et al. [23] present a dynamic programming heuristic for combined vehicle routing and driver scheduling under full EU social legislation, showing that the flexibility of complex break rules yields substantial cost reductions. Prescott-Gagnon et al. [35] embed EU driver rules as resources in a column-generation framework for the VRP with time windows, treating the TDSP as a pricing subproblem. Rancourt et al. [36] address long-haul routing under US HoS with tabu search, confirming that the choice of embedded scheduling algorithm significantly impacts solution quality. Goel and Vidal [19] compare EU, US, Canadian and Australian HoS regulations through optimization, finding that EU rules achieve the highest road safety at a moderate cost premium. Exact branch-and-price algorithms for the combined VRTDSP are developed by Goel and Irnich [18], which is the first with proven optimality guarantees. This method is subsequently accelerated by Tilk and Goel [39] via bidirectional labeling. The most comprehensive EU treatment to date is Garaix et al. [14], who develop a label-setting algorithm for the weekly TDSP and are the first to model the night-work rule, which limits shift working time to ten hours when night work is performed.

► COMMENT

This subsection is in good shape. One addition is worth making: none of these works treats the TDSP under *uncertainty*. Saying so explicitly here, in one sentence, is what earns the paper the right to Section 6.

► SUGGESTED NEW TEXT

Suggested closing sentence: “Across this literature the driver schedule is computed once, against deterministic travel times. To our knowledge no exact EU TDSP formulation has been embedded in a sequential decision framework in which leg durations are revealed as the route is executed, which is the setting of the present paper.”

2.2. The electric vehicle routing problem

The electric vehicle routing problem (EVRP) extends classical vehicle routing to account for the limited driving range of battery electric vehicles and the need for en-route recharging at charging stations. For a comprehensive survey we refer to Erdelić and Carić [9]. The

foundational formulation is the green VRP of Erdoğan and Miller-Hooks [10], which routes alternative-fuel vehicles through refueling stations with full recharging. Exact branch-price-and-cut algorithms for the EVRPTW, covering four variants from single full recharges to multiple partial recharges, are developed by Desaulniers et al. [7], who establish that allowing both multiple and partial recharges reduces routing cost and fleet size.

Partial recharging is studied by Felipe et al. [12] for a multi-technology green VRP and it is shown by Keskin and Çatay [21] to substantially reduce route duration. Depot-level charge scheduling over multiple days, accounting for time-dependent energy prices and battery degradation, is studied by Pelletier et al. [31]. The charging function for battery follows a concave nonlinear profile due to the constant-current/constant-voltage physics of lithium-ion cells. While most work simplify this assumption by assuming a linear charging function, it may also be linearized through the use of a piecewise-linear (PWL) approximation [25].

Uncertainty at charging stations has received growing attention: Keskin et al. [22] address the EVRPTW with stochastic waiting times due to limited charger availability, showing that queuing uncertainty significantly affects routing decisions. Energy consumption uncertainty is addressed by Pelletier et al. [32] using a robust optimization framework, demonstrating that robust solutions come at only a modest cost increase. The bulk of EVRP research targets light-duty vehicles in urban or regional settings. Battery electric heavy-duty trucks (BEHDTs) differ qualitatively: much larger energy demands per leg, reliance on high-power fast charging, and binding HoS constraints that govern both when the driver may be on duty and how charging time interacts with mandatory break requirements.

2.3. Combining TDSP and EVRP

Despite the practical importance of long-haul BET operations, the literature combining HoS regulation with battery energy management in a single optimization model is sparse. Bai et al. [3] study the problem of jointly determining optimal charging and rest decisions for an electric truck on a fixed pre-planned route under HoS constraints. They formulate a bilinear MIP minimizing total cost (charging plus rest time) and develop a rollout-based approximate solver that scales linearly with the number of stations, validated on Swedish road network data.

Brockmann and Guajardo [6] develop a mathematical programming model that synchronizes driver break times with truck charging events under EU HoS, validated on real-world data from northwest Germany. Their key finding is that mandatory breaks create natural charging windows sufficient to match diesel route durations. Wan et al. [41] study long-haul truck charging planning with time and energy flexibility, proposing a mixed-integer model for opportunity charging at routine stops. Su et al. [38] introduce a stage-expanded graph for minimizing the carbon footprint of long-haul heavy-duty e-truck trips, simultaneously optimizing path, speed, and charging, and show that the optimized charging strategy compounds the emissions benefit of electrification. Vidotte Plaza et al. [40] are the first to jointly optimize charging infrastructure siting and HoS-compliant driver scheduling for long-haul eHDT fleets, though their focus is on strategic network design rather than operational online scheduling. Finally, Niyomphon et al. [28] propose a chance-constrained stochastic MILP

for electric freight on fixed long-haul routes with partial recharging and uncertain freight demand, but without HoS constraints.

► TO DO

Brockmann and Guajardo [6] is the paper this work most directly engages with, and Section 8.3.3 produces a quantitative *qualification* of their headline result. Flag that here — one sentence, forward-referencing Section 8.3.3 — so that a referee reading the review already knows the paper has something to say back to the closest prior work, rather than discovering it 25 pages later.

2.4. Uncertainty in long-haul freight transport

Travel time uncertainty in long-haul road freight creates two risks: a HoS violation when accumulated driving or working time exceeds its limit before a scheduled break, and battery depletion when a leg ~~takes longer than nominal or use more energy than planned~~ **is driven faster than nominal and therefore consumes more energy than planned**. While stochastic EVRP models have addressed energy consumption uncertainty [32] and queuing uncertainty at chargers [22], travel time uncertainty and its joint effect on HoS compliance and SOC feasibility remains largely unexplored.

► CODE vs. PAPER MISMATCH

This is the single most consequential factual error in the manuscript. Under the paper’s own energy model $ECR(v) = A/v + B + Cv^2$ with the calibrated coefficients in `src/settings.py`, the consumption rate has its *minimum* at $v \approx 61$ km/h and is increasing over the whole realizable band $[50, 90]$ km/h:

ξ	0.889 (fastest)	1.00 (nominal)	1.20	1.60 (slowest)
speed (km/h)	90	80	67	50
energy, 25 km leg (kWh)	29.40	27.49	26.04	26.67

A delayed leg therefore consumes *less* energy than planned, not more. Stranding risk comes from the *fast* tail, HoS risk from the *slow* tail. Section 6.5 already gets this right (that is exactly why $\mathcal{U}_T$ carries two independent budgets); Sections 1, 2.4 and 6.4 do not.

► SUGGESTED NEW TEXT

This is worth more than a correction — there is a result hiding in it. Suggested addition to Section 8.2 or to the conclusion:

“Because the energy deviation induced by travel-time uncertainty is bounded at +7% per leg over the whole support, travel-time noise alone cannot account for the 12.6% stranding rate of the two-stage plan. That rate is therefore evidence of a structurally mis-placed charging plan that duration-only recourse cannot repair, not of an energy shock the plan could not have anticipated.”

The framework of this type of problem where decisions have to be made sequentially, and adjusted based on realized uncertainty, fits well with the rolling horizon methodology, especially when the multi stage optimization problem would be too complex to solve. The

rolling-horizon paradigm is a well-established technique for multi-period optimization under gradual information revelation [15]. The broader literature on dynamic vehicle routing [33] covers stochastic programming, robust optimization, and online policies, and classifies our problem as one with a fully known route structure but sequentially revealed stochastic leg costs.

2.5. Contribution

Table 1 positions the present work against the most closely related papers across ~~eight~~**six** dimensions. This paper is the first to jointly model full EU HoS (including split-break sequencing, shift working-time limits, multiple rest types, and charging-break interactions), partial fast-charging with a nonlinear PWL charging curve, multiple customer stops with soft time windows on a fixed long-haul route, and stochastic travel times.

~~CHECK AGAIN TABLE~~

► CORRECTION / REWRITE

The count was wrong (the table has six comparison columns, not eight) and, more importantly, **the table has no column for the paper’s actual contribution**. The manuscript claims the decision-architecture comparison as the methodological contribution, but Table 1 cannot express it: it has no “uncertainty handling” and no “benchmark” dimension. Two columns have been added below. A referee reading only the table would currently conclude the contribution is “EU HoS + non-linear partial charging”, which Brockmann and Guajardo [6] already largely has.

► TO DO

The dynamic-vehicle-routing and rolling-horizon strand cited in Section 2.4 has *no row* in Table 1. Add at least one representative rolling-horizon EVRP entry, otherwise the table reads as though it were assembled for a purely deterministic paper and the “uncertainty handling” column has nothing to contrast against.

3. Problem Definition

This paper studies the problem of scheduling driver activities for long-distance freight transport with a battery electric truck (BET). The schedule must simultaneously satisfy Hours-of-Service (HoS) regulations and battery energy constraints. Given a predefined sequence of customer visits, the objective is to minimize the total route duration by determining the optimal timing and locations of mandatory breaks, daily rest periods, and battery charging operations.

► SUGGESTED NEW TEXT

Missing: an explicit assumptions subsection. TR-E expects one, and here it is genuinely needed — several of the assumptions below are currently discoverable only by reading the code, and at least two of them (no idle waiting, deterministic queue) do real work in the results. Suggested text, to be placed as §3.1:

3.1 Modelling assumptions. The following assumptions delimit the problem.

Table 1: Positioning of the present work in the literature. **Two columns added: how uncertainty is handled, and whether the study reports a performance bound.**

Reference	HoS	Charging	Route length	Fixed	TW	Uncertainty	Uncert. handling	Bound
Goel and Irnich 18	EU	–	Long-haul	–	Hard	–	deterministic	exact
Garaix et al. 14	EU	–	Long-haul	✓	Hard	–	deterministic	exact
Montoya et al. 25	–	Partial + NL	Regional	–	Hard	–	deterministic	exact
Keskin et al. 22	–	Partial + L	Long-haul	–	Hard	CS queues	simulation heuristic	–
Niyomphon et al. 28	–	Partial + L	Long-haul	–	–	Demand	chance constraint	–
Bai et al. 3	EU	Partial+L	Day	✓	–	–	rollout (determ.)	–
Brockmann and Guajardo 6	EU	Partial+NL	Day	✓	–	–	deterministic	exact
Vidotte Plaza et al. 40	Brazil	Full+L	Long-haul	–	Hard	–	deterministic	–
This paper	EU	Partial+NL	Long-haul	✓	Soft	Travel time	5 architectures: box RO, budgeted RO, 2-stage SP, rolling horizon, greedy	hindsight oracle

Charging: L/NL: linear/non-linear charging function. **TW:** customer time windows.

- (A1) *Fixed sequence.* The order of customer visits is given and not a decision. Routing is therefore out of scope; the problem is a scheduling problem on a line.
- (A2) *Single vehicle, single trip.* One truck executes one route between two weekly rests. Fleet-level effects and depot slow charging are excluded.
- (A3) *No idle waiting.* Service starts on arrival and the vehicle never waits voluntarily; the only way to delay a departure is to declare a break or a rest. Early arrival at a customer is therefore penalised, not absorbed.
- (A4) *Daily HoS provisions only.* We model the daily provisions of Regulation (EC) No 561/2006 and Directive 2002/15/EC. Weekly-level accounting — weekly rests, the 56 h weekly driving cap and the 60 h working week — is out of scope; the reduced-rest and extended-driving budgets $\bar{\rho}$ and $\bar{\epsilon}$ are applied per route, which is exact when the route is completed between two weekly rests.
- (A5) *Energy depends on leg-average speed.* Consumption follows a speed dependent rate $\text{ECR}(v)$ evaluated at the realized leg-average speed. Grade, payload variation, auxiliary loads and the extra consumption of stop-and-go traffic at a given average speed are not represented.
- (A6) *Charger access is deterministic.* The access delay Q_i at each charging station is a known parameter of the instance, identical for every method. Charger availability, reservation and endogenous queueing are out of scope.
- (A7) *Makespan objective.* The objective is route duration plus a fixed service-level penalty per missed window. Energy price, driver wage and charging tariffs are not priced.

3.1. Route and Customers

The route connects a sequence of customer locations via road segments whose lengths are known in advance. Travel on each segment is characterized by an average speed. In practice, however, traffic conditions introduce uncertainty in the actual travel duration: this is explicitly accounted for in the model. The total route duration is the sum of ~~four~~**five** components: driving time accumulated while the vehicle is moving between locations, service time for loading or unloading operations, charging time at charging stations, **station access and manoeuvring overheads**, and legally required interruptions such as mandatory breaks and daily rest periods.

► COMMENT

The manoeuvring/queue/repositioning terms are not a rounding detail: the diesel decomposition in Section 8.3.3 shows they consume roughly half of the break-charge coupling credit. They deserve to be named in the list of components.

The sequence of customers to be visited is fixed and known prior to departure. At each customer location, the driver performs a loading or unloading operation whose duration is known. At each customer, soft time windows are considered. Service begins immediately upon arrival, whether or not the window is open: distribution terminals do not turn a truck away, but an arrival outside the announced window, either early or late, disrupts terminal planning and incurs a fixed service-level penalty, identical for early and late arrivals. Any break or rest taken at a customer stop occurs after service.

► TO DO

Justify β . The paper’s central uncertainty finding is “uncertainty is paid in windows, not in driving time”, and the size of that finding is proportional to β , which Table 5 sets to 1 h with no argument. `settings.py` already carries `BETA_TW_CLASSES = (1, 2, 5)` and the sweep was cut from scope. Two options, in order of preference: (i) run the $\beta \in \{1, 2, 5\}$ sweep on the reduced set and report it as a robustness check; or (ii) promote the *duration-only* gap to the headline column of Table 8 and relegate the penalised gap to parentheses — the reverse of the current presentation — which removes the paper’s exposure to the choice of β at zero computational cost.

3.2. Charging operations

The BET has a battery of fixed capacity (kWh). Energy consumption is assumed to be dependent of the distance traveled and the vehicle speed, yielding a specific consumption rate expressed in kWh/km. At each CS stop, the driver may charge the battery either partially or fully, and multiple charging stops are permitted throughout the route. Charging is not allowed during customer service operations.

Only high-power fast charging is available along the route (overnight slow charging such as in depot is not considered). The charging process is modeled by a piecewise linear approximation of the nonlinear charge-rate curve. ~~Queuing at a CS is modeled as a known waiting time at each CS.~~ **Charger access delay is represented as a known, instance-specific parameter Q_i incurred whenever the truck charges at station i . It is drawn once at instance generation and is visible to every method, so it is a deterministic cost of using a station rather than a source of uncertainty; endogenous queueing and charger unavailability are out of scope.**

► CODE vs. PAPER MISMATCH

The earlier manuscript wording (“Queuing at a CS is modeled explicitly”) was misleading and has already been softened in your source. The version above states what `settings.py` actually does: $Q_i \sim \text{LogNormal}$ drawn once per instance at generation time, never re-drawn, never revealed on arrival. Worth being explicit, because *Greedy’s* queue-avoidance rule would otherwise look like privileged information — it is not, since Q_i enters the departure-time constraint of every method’s model.

Finally, a sufficiently long charging stop may simultaneously satisfy a mandatory break requirement. Specifically, a charging event counts as a driving break provided its duration meets the applicable threshold: at least 15 minutes for a first split break, at least 30 minutes for a second split break, or at least 45 minutes for a full break, as defined by the HoS regulations described below.

► TO DO

Give the legal basis for the “charging counts as a break” rule in a footnote (Art. 7 of Reg. 561/2006 requires only that the driver not drive and be free to dispose of the time; supervising a charge that requires no intervention is generally read as compatible). This is the load-bearing modelling assumption of the whole paper and it is currently asserted without authority.

3.3. HoS Regulations

The scheduling problem is subject to European HoS regulations given by Regulation (EC) No 561/2006 (driving times, breaks and rests) and Directive 2002/15/EC (working time). We model the *daily* provisions of both instruments. Weekly-level accounting — the weekly rest, the 56 h weekly driving limit and the 60 h weekly working-time cap — is outside the scope of this study; the reduced-rest and extended-driving allowances are consequently budgeted per route, which is exact under the assumption that each route is executed between two weekly rests. Two accumulation counters are maintained: driving time, which counts only periods when the vehicle is in motion, and working time, which additionally includes customer service, charger access and maneuvering, and charging time not covered by a simultaneous break.

► CODE vs. PAPER MISMATCH

Blocking. MILP.py lines 1241–1250 state that the weekly cap is “OUT OF PROBLEM SCOPE (2026-07-29) ...no model enforces it and the simulator does not treat a breach as infeasible”. But constraint (was 54) appears in Section 4, T_{60}^{wk} appears in Table 2, and Section 7.2 claims that “Large instances approach the weekly driving cap” — a cap that is not modelled. All three must go; the red sentence above replaces them.

► SUGGESTED NEW TEXT

The simulator *does* accumulate realized weekly working time and records a breach as a diagnostic (`metrics.weekly_cap_exceeded`). Turn the hole into a reported limitation: “On the long-route class the modelled schedules span four to five duty cycles, at which point the weekly provisions we do not model would begin to bind. We record realized weekly working time as a diagnostic and report the share of long-route schedules that would breach the 60 h cap, as a measure of how far the daily-only model can be trusted on that class.” [TO DO: compute that share from the stored runs — it is already logged.]

Regarding mandatory breaks, a driver may not accumulate more than 4.5 hours of continuous driving without taking a full break of at least 45 minutes. This full break may be replaced by a split sequence consisting of a first period of at least 15 minutes (but less than 45 minutes) followed, in that order, by a second period of at least 30 minutes.

Regarding shift-level constraints, the maximum permitted driving time per shift is 9 hours, extendable to 10 hours on no more than two occasions per week. A daily rest of at least 11 h must separate consecutive shifts, and it may be reduced to 9 h at most three times between weekly rests. As a consequence of these rest requirements, the sum of working time, break time, and any other on-duty periods within a 24-hour window is bounded by 13 hours under a full rest, or by 15 hours when the rest is reduced to 9 hours.

► COMMENT

This last sentence encodes a small but genuine modelling result — that the “daily rest completed within 24 h” requirement is *equivalent* to a bound on the shift spread h_i . It is currently buried. Give it a named remark in Section 4 (see the annotation at (55)); it is one of the few places where the formulation does something an existing paper does not simply already do.

Charging during rests is treated as sequential: the truck must vacate the charging bay before the rest begins, and such charging time counts as work.

While charging is only possible at charging stations, breaks and rests may be taken at customer stops, charging stations, and dedicated rest-area nodes inserted along legs.

4. Deterministic Mathematical Model

We now present the mathematical formulation for the deterministic problem. Tables 2 and 3 present all notation used in the model formulation. Figure \ref{fig:2} presents a possible solution of the problem.

► CORRECTION / REWRITE

Broken cross-reference in the source (`\ref{}` with an empty argument), which compiled to “Figure ??”. Repaired above to point at the example-solution figure.

► TO DO

Section 4 is very long and formula-dense (60 numbered equations plus two appendices). For TR-E, consider moving the piecewise-linear block (13)–(22) and the split-break propagation (39)–(41) to the appendix, and keeping in the body only what is genuinely new: the charge/break credit (24)–(45) and the shift-spread construction (54)–(56). Those two are the formulation’s contribution; the rest is standard and its bulk currently hides them.

► COMMENT

Three notation changes above. (i) $\mathcal{T}_{\min}$ was listed but never given a value anywhere in the paper; the code fixes it at 15 min (MILP.py: `tauc[i] >= 0.25*y[i]`). (ii) The symbols $T_{sh1}^{wk}, T_{sh2}^{wk}$ in the original table are the *shift spread* bounds and are called T_1^{spr}, T_2^{spr} everywhere in the constraint section. Two names for one parameter; unified to T^{spr} . (iii) $\bar{\rho}$ and $\bar{\epsilon}$ are described as weekly budgets but are enforced once per route (37), (52), which is exact only under assumption A4.

4.1. Sets and Parameters

Let the route consist of a fixed sequence of stops indexed by $i \in \mathcal{I} = \{0, 1, \dots, N\}$, where stop 0 is the origin and stop N is the destination. The set of customer stops is $\mathcal{C} \subseteq \mathcal{I}$, and the set of CS stops is $\mathcal{K} \subseteq \mathcal{I}$. Additionally, rest areas $\mathcal{L} \subseteq \mathcal{I}$ are placed along the route and are treated as charging stations but with charging disabled. The set of non-customer stops $\mathcal{S}$ includes both charging stations and rest areas, such that $\mathcal{C} \cap \mathcal{K} \cap \mathcal{L} = \emptyset$ and $\mathcal{K} \cup \mathcal{L} = \mathcal{S}$. The model structure is similar to that of Bai et al. [3], adapted here to a single-vehicle fixed-route setting with multiple customer stops. The nonlinear charging curve is approximated by a piecewise linear function defined by $R + 1$ breakpoints, following Montoya et al. [25].

4.2. Objective Function

The objective minimizes the arrival time at the destination plus a fixed penalty β for every customer whose service starts outside its window.

$$\min \quad t_N^a + \beta \sum_{i \in \mathcal{C}} \delta_i \tag{1}$$

Table 2: Notation summary.

Sets	
$\mathcal{I}$	Set of possible stops, ordered from 0=origin to N=destination.
$\mathcal{C}$	Set of customers $\mathcal{C} \subseteq \mathcal{I}$.
$\mathcal{K}$	Set of charging stations $\mathcal{K} \subseteq \mathcal{I}$.
$\mathcal{L}$	Set of rest areas $\mathcal{L} \subseteq \mathcal{I}$, $\mathcal{C} \cap \mathcal{K} \cap \mathcal{L} = \emptyset$.
$\mathcal{R}$	Set of breakpoints of the piecewise-linear charging function.
Parameters	
D_i	Travel time from stop i to stop $i + 1$.
E_i	Energy consumption between stop i and stop $i + 1$.
E^{cap}	Battery capacity.
$E^{\min}$	Minimum SOC.
E^0	Initial SOC.
$\bar{E}_r$	SOC level at breakpoint $r \in \mathcal{R}$ of the charging fct, ($\bar{E}_0 = 0$, $\bar{E}_R = E^{cap}$).
$\mathcal{T}_r$	Time to charge from empty to $\bar{E}_r$, ($\mathcal{T}_0 = 0$).
$\mathcal{T}_{\min}$	Minimum charging time at a CS (= 15 min).
S_i	Service time at customer $i \in \mathcal{C}$.
Q_i	Queue time at CS stop $i \in \mathcal{K}$ if charging occurs. Zero if $y_i = 0$.
M_i^{stop}	Maneuver time at CS stop $i \in \mathcal{S}$ if any activity occurs.
M_i^{seq}	Repositioning time at stop $i \in \mathcal{K}$ when charging and a break or rest are declared sequentially, i.e. the truck must vacate the charging bay before the break begins.
W_i^{Ha}, W_i^{Hf}	Service time window at customer $i \in \mathcal{C}$.
β	Fixed out-of-window service penalty (h).
T_{b45}^{brk}	Full break duration (= 45 min).
T_{b15}^{brk}	Min first split break (= 15 min).
T_{b30}^{brk}	Min second split break (= 30 min).
T_{r1}^{rst}	Minimum daily rest duration (= 11 h).
T_{r2}^{rst}	Reduced daily rest duration (exception 1) (= 9 h at most 3×/week).
$\bar{\rho}$	Weekly n Number of allowed reduced rests per route (= 3).
T_{cons}^{drv}	Max consecutive driving time (= 4.5 h).
T_{shift1}^{drv}	Max shift driving time (= 9 h).
T_{shift2}^{drv}	Increased shift driving time (exception 2) (= 10 h at most 2×/week).
$\bar{\epsilon}$	Weekly n Number of allowed extended driving shifts per route (= 2).
T_1^{spr}	Max shift spread under a regular daily rest (= 13 h).
T_2^{spr}	Max shift spread under a reduced daily rest (= 15 h).

[CODE:

T_{60}^{wk} is
used in
constraint
(was 54)
but declared
nowhere in
this table
— and the
constraint
itself is
not imple-
mented.

Table 3: Variables notation summary.

Variables	
$t_i^a, t_i^d \geq 0$	arrival/departure time at stop $i \in \mathcal{I}$.
$\delta_i \in \{0, 1\}$	1 if service at customer $i \in \mathcal{C}$ starts outside its window.
$y_i \in \{0, 1\}$	1 if the truck charges at stop $i \in \mathcal{K}$.
$\tau_i^c \geq 0$	charging duration at stop $i \in \mathcal{K}$.
$e_i^a, e_i^d \geq 0$	SOC at arrival/departure at CS at stop $i \in \mathcal{I}$.
$\lambda_{i,r}^a, \lambda_{i,r}^d \geq 0$	PWL weight for encoding $e_i^a, e_i^d, i \in \mathcal{K}, r \in \mathcal{R}$.
$\mu_{i,r}^a, \mu_{i,r}^d \in \{0, 1\}$	1 if e_i^a, e_i^d lies in segment r of the PWL curve.
$c_i^d \geq 0$	accumulated consecutive driving time at arrival at stop i (h).
$s_i^d \geq 0$	accumulated shift driving time at arrival at stop i (h).
$s_i^w \geq 0$	accumulated shift working time at arrival at stop i (h).
$h_i \geq 0$	Elapsed time since end of last daily rest (h).
$x_i^{b_{45}}, x_i^{b_{15}}, x_i^{b_{30}} \in \{0, 1\}$	break type indicators at stop i .
$\phi_i \in \{0, 1\}$	1 if the first split break has been taken since the last driving reset.
$\tau_i^b \geq 0$	Break time beyond charging at stop i .
$\hat{\tau}_i^b \geq 0$	Break duration credited for HoS purposes.
$g_i \geq 0$	Charging time credited as break at stop i .
$\sigma_i \in \{0, 1\}$	1 if CS i operates in sequential mode (no charging/break synchronization).
$\rho_i^{r1}, \rho_i^{r2} \in \{0, 1\}$	rest type indicators at stop i .
$\tau_i^r \geq 0$	rest duration at stop i .
$v_i \in \{0, 1\}$	1 if any activity (charge/break/rest) occurs at stop i .
$z_i, q_i \in \{0, 1\}$	extended shift flag; extension consumed at a rest.
$w_i \geq 0$	idle wait at stop i ; <i>fixed to zero</i> in the base model (assumption A3), released only for the benchmark parity runs of Section 8.1.

Abbreviations: HoS = Hours-of-service, CS = Charging station

FIGURE NOT AVAILABLE IN THE REPOSITORY

[figures/routeExample.png](#)

Figure 1 of the manuscript — the corridor schematic with origin, CS triangles, customer diamonds and rest-area ticks. Present on Overleaf, absent from the repository, so it could not be re-typeset here.

Figure 1: ~~Enter Caption~~ Structure of a benchmark corridor. Charging stations (triangles) are placed on a 60 km grid with ± 19 km jitter; rest-area nodes (ticks) subdivide any leg longer than 25 km and permit a break or rest but no charging; customers (diamonds) are drawn around cluster centres at 25%, 50% and 70% of the corridor. The instance shown is a short-route case: 900 km, three customers.

► COMMENT

A referee at TR-E will ask why the objective is makespan rather than cost. The answer is defensible — on a fixed route with a fixed vehicle, driver time is the dominant marginal cost and asset utilisation scales with makespan — but it has to be *said*, in one or two sentences, right here. Consider also converting the headline result once into money in the conclusion (“+9.4% on a 53 h medium route is +5 h”) so the magnitude is legible to a logistics audience.

4.3. Constraints

In order to simplify the following constraints, we define additional variables as such: $\rho_i = \rho_i^{r1} + \rho_i^{r2} \in \{0, 1\}$ is the rest indicator at a stop, $x_i = x_i^{b45} + x_i^{b15} + x_i^{b30} \in \{0, 1\}$ is the break indicator at a stop, $r_i = x_i^{b45} + x_i^{b30} + \rho_i^{r1} + \rho_i^{r2} \in \{0, 1\}$ is the reset indicator of consecutive drive time at a stop. We further write $o_i = t_i^d - t_i^a - \tau_i^r$ for the on-duty (non-rest) time spent at stop i , used in the worktime shift counter.

4.3.1. Time propagation

Constraint (2) states that the arrival time at each stop equals the departure time from the previous stop plus the leg travel time. The departure time from a stop depends on the activities undertaken there. At a customer stop, constraint (3) accounts for mandatory service time S_i and any additional break or rest duration. At a CS stop, constraint (4) additionally includes a queue time $Q_i y_i$, a charging duration τ_i^c , a maneuver time M_i^{stop} to get to/leave the CS in case we stop there for any activity, a maneuver time M_i^{seq} if we first charge and then sequentially take a break or rest at the CS. Constraints (5)–(6) activate the out-of-window indicator δ_i whenever service starts before the window opens or after it closes. Since service starts at arrival, the window applies to t_i^a . **Note that (3) and (4) contain no idle-wait term: the schedule cannot absorb an early arrival by waiting (assumption A3), only by declaring a break or a rest.**

$$t_{i+1}^a = t_i^d + D_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (2)$$

$$t_i^d = t_i^a + S_i + \tau_i^b + \tau_i^r, \quad i \in \mathcal{C} \quad (3)$$

$$t_i^d = t_i^a + M_i^{stop} v_i + Q_i y_i + \tau_i^c + M_i^{seq} \sigma_i + \tau_i^b + \tau_i^r, \quad i \in \mathcal{S} \quad (4)$$

$$t_i^a \leq W_i^f + M \delta_i, \quad i \in \mathcal{C} \quad (5)$$

$$t_i^a \geq W_i^a - M \delta_i, \quad i \in \mathcal{C} \quad (6)$$

4.3.2. Battery state-of-charge

Constraint (7) propagates the SOC between stops according to the energy consumed on each leg. Constraints (8)–(9) enforce that charging can only occur at designated CS stops and can never decrease the SOC. Constraints (10)–(11) keep the SOC within the feasible battery operating range at all times. Constraint (12) ensures that a positive charging duration can only be incurred when the binary y_i is active, and that in case of charging, we charge at least as much as the minimum charging time allowed.

$$e_{i+1}^a = e_i^d - E_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (7)$$

$$e_i^d = e_i^a, \quad i \in \mathcal{C} \cup \mathcal{L} \quad (8)$$

$$e_i^d \geq e_i^a, \quad i \in \mathcal{K} \quad (9)$$

$$e_i^a \geq E^{\min}, \quad i \in \mathcal{I} \quad (10)$$

$$e_i^d \leq E^{\text{cap}}, \quad i \in \mathcal{I} \quad (11)$$

$$\mathcal{T}_{\min} y_i \leq \tau_i^c \leq \mathcal{T}_R y_i, \quad i \in \mathcal{K} \quad (12)$$

► COMMENT

$E^{\min} = 0.2 E^{\text{cap}}$ is a modelling choice (range anxiety plus battery health) rather than a physical limit, and it costs 100 kWh of usable capacity on a 500 kWh pack — roughly one 90 km leg. It is worth one justifying sentence in Section 7, with a citation, since the stranding rates reported in Table 9 are all measured against this threshold and not against an empty battery.

4.3.3. Piecewise-Linear Charging Function

Following Montoya et al. [25], we linearize the nonlinear charging function using a convex-combination formulation. Constraints (13)–(14) express both the arrival and departure SOC at each CS stop as convex combinations of the $R + 1$ PWL breakpoints. The charging duration is obtained in constraint (15) as the difference in cumulative charge time between the departure and arrival SOC levels. Constraints (16)–(17) impose the convex-combination normalization condition. Constraints (18)–(19) enforce that exactly one PWL segment is active for each SOC level. Finally, the SOS2 adjacency constraints (20)–(22) restrict non-zero weights to at most two consecutive breakpoints, for both arrival ($\nu = a$) and departure ($\nu = d$) SOC levels.

$$e_i^a = \sum_{r \in \mathcal{R}} \lambda_{i,r}^a \bar{E}_r, \quad i \in \mathcal{K} \quad (13)$$

$$e_i^d = \sum_{r \in \mathcal{R}} \lambda_{i,r}^d \bar{E}_r, \quad i \in \mathcal{K} \quad (14)$$

$$\tau_i^c = \sum_{r \in \mathcal{R}} \lambda_{i,r}^d \mathcal{T}_r - \sum_{r \in \mathcal{R}} \lambda_{i,r}^a \mathcal{T}_r, \quad i \in \mathcal{K} \quad (15)$$

$$\sum_{r \in \mathcal{R}} \lambda_{i,r}^a = 1, \quad i \in \mathcal{K} \quad (16)$$

$$\sum_{r \in \mathcal{R}} \lambda_{i,r}^d = 1, \quad i \in \mathcal{K} \quad (17)$$

$$\sum_{r=1}^R \mu_{i,r}^a = 1, \quad i \in \mathcal{K} \quad (18)$$

$$\sum_{r=1}^R \mu_{i,r}^d = 1, \quad i \in \mathcal{K} \quad (19)$$

$$\lambda_{i,0}^\nu \leq \mu_{i,1}^\nu, \quad (20)$$

$$\lambda_{i,r}^\nu \leq \mu_{i,r}^\nu + \mu_{i,r+1}^\nu, \quad r = 1, \dots, R-1, \quad (21)$$

$$\lambda_{i,R}^\nu \leq \mu_{i,R}^\nu, \quad (22)$$

for all $i \in \mathcal{K}$ and $\nu \in \{a, d\}$.

4.3.4. Break and Rest Decisions

Constraint (23) ensures that at most one type of break or rest activity is declared at each stop.

$$x_i + \rho_i \leq 1, \quad i \in \mathcal{I} \quad (23)$$

At a CS stop, charging time may qualify toward HoS break requirements only when a break is declared and runs in parallel with charging ($x_i = 1, \sigma_i = 0$). Let g_i denote the credited charging time. Then constraints (24)–(27) force $g_i = \tau_i^c$ exactly in that case and $g_i = 0$ otherwise. The effective break duration $\hat{\tau}_i^b$ is then the credited charging time plus any additional break time (28), and the additional break time alone at customer and rest-area stops (29).

$$g_i \leq \tau_i^c, \quad i \in \mathcal{K} \quad (24)$$

$$g_i \leq \mathcal{T}_R(1 - \sigma_i), \quad i \in \mathcal{K} \quad (25)$$

$$g_i \leq \mathcal{T}_R x_i, \quad i \in \mathcal{K} \quad (26)$$

$$g_i \geq \tau_i^c - \mathcal{T}_R \sigma_i - \mathcal{T}_R(1 - x_i), \quad i \in \mathcal{K} \quad (27)$$

$$\hat{\tau}_i^b = g_i + \tau_i^b, \quad i \in \mathcal{K} \quad (28)$$

$$\hat{\tau}_i^b = \tau_i^b, \quad i \in \mathcal{C} \cup \mathcal{L} \quad (29)$$

► COMMENT

(24)–(45) are the core of the paper’s formulation and the one block that no prior model contains in this form. They deserve a short worked sentence — “a 50-minute charge at a station where a full break is declared in parallel discharges the break at zero marginal time cost; a 30-minute charge does not, and σ_i is then forced to one so that the 45-minute break is appended after the charge, at a cost of $M_i^{seq} + 45 \text{ min}$ ” — because the entire diesel comparison of Section 8.3.3 is an accounting of exactly this mechanism.

Constraints (30)–(32) impose the minimum effective break duration for each break type. Similarly, constraints (33)–(34) impose the minimum rest duration for each rest type. Constraints (35)–(36) force the break and rest durations to zero when no corresponding activity is declared. Constraint (37) enforces that the reduced daily rest of 9 h may be used at most $\bar{\rho}$ times ~~per week~~**per route**.

$$\hat{\tau}_i^b \geq T_{45}^{\text{brk}} x_i^{b_{45}}, \quad i \in \mathcal{I} \quad (30)$$

$$\hat{\tau}_i^b \geq T_{15}^{\text{brk}} x_i^{b_{15}}, \quad i \in \mathcal{I} \quad (31)$$

$$\hat{\tau}_i^b \geq T_{30}^{\text{brk}} x_i^{b_{30}}, \quad i \in \mathcal{I} \quad (32)$$

$$\tau_i^r \geq T_1^{\text{rst}} \rho_i^{r_1}, \quad i \in \mathcal{I} \quad (33)$$

$$\tau_i^r \geq T_2^{\text{rst}} \rho_i^{r_2}, \quad i \in \mathcal{I} \quad (34)$$

$$\tau_i^b \leq M x_i, \quad i \in \mathcal{I} \quad (35)$$

$$\tau_i^r \leq M \rho_i, \quad i \in \mathcal{I} \quad (36)$$

$$\sum_{i \in \mathcal{I}} \rho_i^{r_2} \leq \bar{\rho} \quad (37)$$

EU regulation allows the standard 45 min break to be replaced by a split break taken in two parts: a first part of at least 15 min followed by a second part of at least 30 min, in that strict order. The binary ϕ_i tracks whether the first part of a split break has been completed since the last consecutive-driving reset. Constraint (38) ensures the second part can only be declared if the first part was taken at an earlier stop. Constraints (39)–(41) propagate ϕ between stops.

$$x_i^{b_{30}} \leq \phi_i, \quad i \in \mathcal{I} \quad (38)$$

$$\phi_{i+1} \geq \phi_i + x_i^{b_{15}} - r_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (39)$$

$$\phi_{i+1} \leq \phi_i + x_i^{b_{15}}, \quad i \in \mathcal{I} \setminus \{N\} \quad (40)$$

$$\phi_{i+1} \leq 1 - r_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (41)$$

► SUGGESTED NEW TEXT

An unreported result the repository already contains. The split break is an *option* under Art. 7, not an obligation, and `settings.py` carries an `ALLOW_SPLIT_BREAK` switch with a generated `no_split` instance family. Preliminary runs indicate the hindsight optimum uses a split at roughly 38% of instances, yet the makespan effect of forbidding splits sits below the solver’s own MIP gap. That is a clean, quotable, regulation-facing finding: *the split-break provision, which materially complicates every exact EU TDSP formulation, is worth less than half a percent of makespan to an electric truck, because charging already supplies the break.* Suggested as a short subsection in Section 8. [TO DO: re-verify first: 156 oracle caches for this family were found to contain git conflict markers, so the current numbers cannot be trusted as they stand.]

Constraint (42) flags any activity at a stop of $\mathcal{S}$. Sequential mode σ_i is possible only when charging occurs (43). It is forced whenever charging coexists with a rest (44), since the truck must vacate the charging bay before an 11 h or 9 h rest, and it is forced whenever a break is declared that the charge does not fully cover (45). In (45), $b_i = T_{45}^{brk} x_i^{b_{45}} + T_{15}^{brk} x_i^{b_{15}} + T_{30}^{brk} x_i^{b_{30}}$ is the required minimum of the break declared at stop i ; if the credited charge g_i falls short of it, σ_i is forced to 1, and constraint (25) then sets $g_i = 0$ so that the full break sits sequentially after the charge.

$$\frac{1}{2}(y_i + \rho_i + x_i) \leq v_i \leq y_i + \rho_i + x_i, \quad i \in \mathcal{S} \quad (42)$$

$$\sigma_i \leq y_i, \quad i \in \mathcal{K} \quad (43)$$

$$\sigma_i \geq y_i + \rho_i - 1, \quad i \in \mathcal{K} \quad (44)$$

$$\sigma_i \geq \frac{b_i - \tau_i^c}{T_{45}^{brk}} - (1 - y_i), \quad i \in \mathcal{K} \quad (45)$$

4.3.5. Driving regulation

The accumulated consecutive driving time c_i^d increases with each driven leg and is reset to zero by any qualifying break or rest (i.e. whenever $r_i = 1$). Defining the auxiliary $l_i^1 = r_i c_i^d$, constraint (46) expresses the propagation rule. Constraints (A.1)–(A.3) linearize it via a standard big- M reformulation. Constraint (47) enforces the regulatory upper bound.

$$c_{i+1}^d = c_i^d + D_i - l_i^1, \quad i \in \mathcal{I} \setminus \{N\} \quad (46)$$

$$c_i^d \leq T_{\text{cons}}^{\text{drv}}, \quad i \in \mathcal{I} \quad (47)$$

The accumulated shift driving time s_i^d is reset only by a daily rest. Defining $l_i^2 = \rho_i s_i^d$, constraint (48) gives the propagation rule, linearized in (A.4)–(A.6) with big- M $T_{sh2}^{\text{drv}} = 10$ h. Constraint (49) imposes the maximum shift driving time conditional on the extension flag z_i ; (50) lets the flag persist within a shift; (51)–(52) charge one unit of the budget $\bar{\epsilon}$ per extended shift, the term z_N covering an unfinished final shift.

$$s_{i+1}^d = s_i^d + D_i - l_i^2, \quad i \in \mathcal{I} \setminus \{N\} \quad (48)$$

$$s_i^d \leq T_{sh1}^{\text{drv}} + (T_{sh2}^{\text{drv}} - T_{sh1}^{\text{drv}})z_i, \quad i \in \mathcal{I}. \quad (49)$$

$$z_{i+1} \geq z_i - \rho_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (50)$$

$$q_i \geq z_i + \rho_i - 1, \quad i \in \mathcal{I} \quad (51)$$

$$\sum_{i \in \mathcal{I}} q_i + z_N \leq \bar{\epsilon} \quad (52)$$

4.3.6. Working regulation

Working time comprises driving, customer service, CS queuing, maneuvering, and charging time not covered by a synchronized break. The work contribution of charging is simply $\tau_i^c - g_i$, which is linear, so no auxiliary variable is needed. The accumulated shift working time s_i^w is reset only by a daily rest. By defining $l_i^4 = \rho_i s_i^w$, constraints (A.7)–(A.9) provide

the linearization with big-M $T_2^{spr} = 15$ h. Constraint (53) propagates s_i^w . The counter carries no daily cap: the daily-level bound on on-duty time is the shift spread (54)–(56) below, and the binding working-time cap is weekly, imposed by constraint (was 54). The counter is retained for reporting and for the ex-post verification of the Directive’s working-time provisions; no weekly cap is enforced (assumption A4).

$$s_{i+1}^w = s_i^w - l_i^4 + D_i + \begin{cases} M_{i+1}^{stop} v_{i+1} + Q_{i+1} y_{i+1} + (\tau_{i+1}^c - g_{i+1}) + M_{i+1}^{seq} \sigma_{i+1} & \text{if } i+1 \in \mathcal{S}, \\ S_{i+1} & \text{if } i+1 \in \mathcal{C}, \end{cases} \quad i \in \mathcal{I} \setminus \{N\} \quad (53)$$

► CORRECTION / REWRITE

Delete constraint (54). The manuscript’s weekly working-time constraint

$$\sum_i D_i + \sum_{i \in \mathcal{C}} S_i + \sum_{i \in \mathcal{S}} (M_i^{stop} v_i + Q_i y_i + (\tau_i^c - g_i) + M_i^{seq} \sigma_i) \leq T_{60}^{wk} \quad (\text{was 54})$$

is **not in the implementation** and must be removed together with T_{60}^{wk} from Table 2 and the “approach the weekly driving cap” sentence in Section 7.2. Every reported result was produced without it. Leaving it in the paper means every number in Section 8 was generated by a different model from the one described.

The shift spread h_i measures the elapsed time since the end of the last daily rest. Since a daily rest is always the last activity performed at a stop, the elapsed time from the end of the previous rest to the start of a rest at stop i is exactly $h_i + o_i$, where $o_i = t_i^d - t_i^a - \tau_i^r$ is the on-duty stop time. Constraint (54) propagates the spread, with the reset $l_i^5 = \rho_i(h_i + o_i)$ linearized in (A.10)–(A.12). Constraint (55) bounds the pre-rest spread at $T_1^{spr} = 13$ h when the upcoming rest is regular and $T_2^{spr} = 15$ h when it is reduced, implementing the requirement that the daily rest be completed within 24 h of the end of the previous one. Constraint (56) is the global bound for shift length.

$$h_{i+1} = h_i + o_i + D_i - l_i^5, \quad i \in \mathcal{I} \setminus \{N\} \quad (54)$$

$$h_i + o_i \leq T_1^{spr} + (T_2^{spr} - T_1^{spr}) \rho_i^{r2} + T_2^{spr} (1 - \rho_i), \quad i \in \mathcal{I} \quad (55)$$

$$h_i \leq T_2^{spr}, \quad i \in \mathcal{I} \quad (56)$$

► SUGGESTED NEW TEXT

Promote this to a remark. The equivalence being used here is a genuine (if small) modelling contribution and is currently asserted in a subordinate clause. Suggested:

Remark 1. *Enforcing (55) is equivalent to requiring that each daily rest be completed within 24 h of the end of the previous one. A regular rest lasts $T_1^{rst} = 11$ h and a reduced rest $T_2^{rst} = 9$ h; since a rest is always the terminal activity at its stop, the 24 h requirement binds the on-duty spread preceding it at $24 - 11 = 13$ h and $24 - 9 = 15$ h respectively, which is exactly (55). This replaces a rolling-window constraint — which would require time-indexed variables — by a single state variable h_i propagated along the route, and it is what the validation of Section 8.1 confirms against an independent exact method.*

4.3.7. Initial Conditions

Constraints (57)–(59) fix the state at the origin: the truck departs at a given time t^0 , with initial SOC E^0 , and all HoS accumulators and the split-break flag are initialized to zero. No break or rest activity is permitted at the origin or at the destination: $x_0 = \rho_0 = x_N = \rho_N = 0$.

$$t_0^a = t^0, \quad e_0^a = E^0, \quad (57)$$

$$c_0^d = 0, \quad s_0^d = 0, \quad (58)$$

$$s_0^w = 0, \quad \phi_0 = 0. \quad (59)$$

FIGURE NOT AVAILABLE IN THE REPOSITORY

figures/ExampleSolution3.png

Figure 2 of the manuscript — the four-lane concept solution (Route / Truck / SOC / HoS / Driver). Generated by `src/plot/concept_solution_pptx.py`; on Overleaf, not in the repository.

Figure 2: ~~Enter Caption~~A feasible schedule on a 950 km corridor with three customers. The four panels share a common time axis. *Route*: the stops used by the solution. *Truck*: driving, parked, and the shaded intervals during which the vehicle occupies a charging bay. *SOC*: state of charge between $E^{\min}$ (red) and E^{cap} . *HoS*: the consecutive-driving and shift-driving accumulators against their limits. *Driver*: the corresponding driver activity. The schedule illustrates the three charge/break regimes of (24)–(45): at 284 km a 45-min break runs *in parallel* with the charge and is discharged at no time cost ($g_i = \tau_i^c$); at 497 km a 9 h reduced rest forces $\sigma_i = 1$, so the truck vacates the bay and the charge is not credited; at 540 km and 779 km a 15-min and a 30-min charge discharge the two halves of a split break.

► COMMENT

Figure 2 is the most communicative object in the paper and currently carries no caption at all. A reader who understands this figure does not need to read (24)–(45) to follow Section 8.3.3; a reader who does not will struggle throughout. Consider moving it *before* the model, as a motivating figure.

5. Complexity of the problem [Draft section — decide whether to keep]

► TO DO

Decision required. The manuscript carries a complexity section written entirely in red, containing (i) two sub-problem structure conjectures with the proofs marked “TBD”, (ii) a set of bullet-point reading notes on Garaix et al. [14], Drexler and Prescott-Gagnon [8] and Xu et al. [42], and (iii) an unfinished discussion of how the two sub-problems combine.

For TR-E my recommendation is to **cut it**, and to relocate the one part that is genuinely useful — the observation that with charging stops fixed the optimal policy is to charge just enough to reach the next active station, by convexity and monotonicity of the PWL charge-time function $T(\cdot)$ — into Section 4 as a two-sentence structural remark, which additionally

motivates valid inequality (B.2). TR-E is not a complexity venue, the NP-hardness of the surrounding TDSP is already *conjectured* rather than proven in the literature you cite, and a section that visibly ends in “TBD” costs more credibility than the result would buy. Keep it only if you intend to complete both proofs.

► SUGGESTED NEW TEXT

If cut, the salvaged remark reads roughly: “**Remark 2.** For a fixed charging pattern y and in the absence of Hours-of-Service constraints, an optimal solution charges at each active station exactly enough to reach the next active station at $E^{\min}$. This follows from the convexity and monotonicity of the cumulative charge-time function $\mathcal{T}(\cdot)$: shifting a unit of energy from a later station to an earlier one moves it to a weakly steeper part of the curve. The optimal charging *pattern*, however, remains combinatorial, since each active station carries a fixed access cost $M_i^{\text{stop}} + Q_i$; valid inequality (B.2) exploits the resulting lower bound on the number of stops.”

6. Solution methods

Introducing uncertainty into a deterministic scheduling model is non-trivial: the optimal schedule depends on travel times and energy consumption that are not yet known at departure. We compare various methods to tackle the problem at hand. First a simulation framework is created in Section 6.1 to track the real life progress of a truck, based on some decisions at each stop and realized uncertainty at each leg. The remainder of this section is organized by *when* a method commits to a decision and *what* information it uses to do so, as summarized in Table 4: Sections 6.2 and 6.3 describe the two adaptive policies, which re-decide at every stop as the trip unfolds; Sections 6.4, 6.5 and 6.6 describe the three anticipative methods, which fix a full schedule before departure. Finally, Section 6.7 introduces the perfect hindsight model.

Table 4: Solution method characteristics. Reordered to match the order of the subsections, and two columns added so the table expresses the taxonomy the section claims to be organized by.

Methodology	§	Decision setting	Travel time info	Committed before departure
Rolling-horizon (LA)	6.2	Online	Scenarios	nothing
Greedy	6.3	Online	Nominal	nothing
Robust opt. (box, RO)	6.4	Offline	Support only	structure + durations
Budgeted robust (ROBU)	6.5	Offline	Budgeted support	structure + durations
2-stage SP	6.6	Offline + recourse	Scenarios	structure
Oracle	6.7	Ex post	Realized values	—

► CORRECTION / REWRITE

Two changes. (i) The original table lists “Greedy ...Average” for the travel-time distribution; the greedy rule uses *nominal* values, and with the feasibility guard disabled (see [CODE:] note in Section 6.2) it uses no distributional information whatsoever. (ii) The original ordering (RO, ROBU, 2SP, RH, Greedy, Oracle) is the reverse of the order in which the subsections appear, which makes the table harder to use than no table.

6.1. Simulation Framework

We decouple the planning problem from the execution problem through a simulation framework, illustrated in Figure 3.

The simulation proceeds stop by stop from origin 0 to destination N . At each stop i , the vehicle has arrived with a known state $\Gamma_i = (t_i^a, e_i^a, c_i^d, s_i^d, s_i^w, h_i, \phi_i, n_i^{r^2}, n_i^\epsilon)$ that summarizes accumulated time, energy, and HoS counters, where $n_i^{r^2}$ and n_i^ϵ count the reduced rests and driving extensions already consumed. A decision method maps this state to an action a_i specifying whether to charge, which break type to take, and which rest type to take. The vehicle executes a_i , departs, and the travel time on leg $i \rightarrow i+1$ is revealed as a random draw, from which the realized energy consumption follows via the ECR curve. The truck state is updated accordingly and the cycle repeats. Early arrival at a customer triggers immediate service, with the out-of-window penalty recorded; **the vehicle cannot wait for a window to open (assumption A3).**

FIGURE NOT AVAILABLE IN THE REPOSITORY

`figures/simulation.png`

Figure 3 of the manuscript — the simulation loop schematic.

Figure 3: The simulation framework. At each stop the realized state Γ_i is passed to a decision selector, which returns an action a_i (charge / break type / rest type). The action is executed, the vehicle departs, and only then is the travel-time multiplier ξ_i of the next leg drawn; the realized energy follows from the ECR curve at the implied speed $L_i/(D_i\xi_i)$. All six methods share this loop and this state representation; they differ only in the box labelled “decision selector”.

The decision methods are described below and they share the same simulation loop and vehicle state representation, only the decision step differs. The travel-time realizations are recorded throughout and fed to the oracle at the end of each run.

Because uncertainty is revealed only after departure, a leg may retroactively cause a violation that no stop-level action can repair: the consecutive-driving counter may exceed 4.5 h mid-leg or the SOC may fall below acceptable level. The simulator records these events and runs with any violation are marked infeasible.

► SUGGESTED NEW TEXT

This paragraph defines the paper’s most important measurement convention and it does so in three lines. It needs two more sentences, because Section 8.2 leans on them heavily:

“Two consequences follow. First, a non-zero violation rate is unavoidable for *any* online policy: a leg whose realized duration exceeds the remaining consecutive-driving budget breaches Article 7 regardless of what was decided at the preceding stop, so the greedy policy’s violation rate is a property of the uncertainty and the corridor, not only of the rule. Second, an infeasible run has no meaningful route duration, so it is excluded from every gap statistic. A method that fails often is therefore scored only on the realizations it survives, and efficiency and reliability must be read together — Table 8 against Table 9 — rather than separately.”

6.2. Rolling-Horizon with look-Ahead policy

We adopt a *rolling-horizon* policy in which a new decision is made at each stop, using only the information available at that point and considering a limited horizon to model the future. Let $i \in \{0, \dots, N - 1\}$ denote the current stop and let Γ_i be the vehicle state at arrival. At each stop the policy executes the following steps, summarized in Figure 4.

All structurally feasible actions $a \in \mathcal{A}_i$ are enumerated. An action specifies the charge binary $y \in \{0, 1\}$ (relevant at CS stops only), a break type $b \in \{\emptyset, b_{45}, b_{15}, b_{30}\}$, and a rest type $r \in \{\emptyset, r_1, r_2\}$. Actions that violate structural feasibility (such as a split-break second part b_{30} without a prior b_{15} , or a reduced rest r_2 when the budget is exhausted) are excluded.

► CODE vs. PAPER MISMATCH

The manuscript’s next sentence — already commented out in your source, and correctly so — claimed that actions are additionally pruned by a worst-case feasibility check. `settings.py` sets `GUARD_QUANTILE = None`, which *disables* that guard: “LA does NO flag-based pruning at all — infeasible actions are exposed by the scenario scores”. Keep it deleted. But say what *is* true, because otherwise the reader is left wondering how LA avoids stranding at all: “No probabilistic feasibility guard is applied: an action that would strand the vehicle is not filtered out a priori but is scored as infeasible in the scenarios that expose it, and is therefore penalised through $f(a)$. This keeps the policy’s information set identical to the greedy policy’s.”

Then, a horizon end-stop $h > i$ is determined by advancing along the route for a nominal duration of T_{hor} hours, based on the rest rules.

For each travel leg between the current stop i and the horizon end-stop h , a set of S travel-time scenarios $\{\xi^{(k)}\}_{k=1}^S$ is drawn independently. Each multiplier $\xi_i^{(k)}$ is sampled from the travel-time distribution of Section 7.1. The corresponding scenario energy $E_i^{(k)}$ is derived from the ECR curve evaluated at the implied speed.

Afterward, for each candidate action $a \in \mathcal{A}_i$ and each scenario k , the sub-problem MILP over the horizon $[i, h]$ is solved with a fixed at the current stop and with travel times $D_i^{(k)} = D_i \xi_i^{(k)}$. The subproblem minimizes arrival time at h , subject to the same SOC, HoS, break, rest, and charging constraints as the full-route model. The score of action a is

$$f(a) = \frac{1}{S} \sum_{k=1}^S t_h^{a,(k)}(a),$$

with infeasible scenarios penalized by a large constant. ~~Alternative aggregations (worst-case, best-case, CVAR) to be considered~~

► COMMENT

The alternative aggregations are *already implemented* (`criterion = "mean" | "worst" | "best" | "cvar"` in `Simulation.py`, CVaR at level 0.8). So this is not an open question but an un-run experiment. Either report it as a one-line robustness check on the reduced instance set, or delete the sentence — but do not ship a paper that says “to be considered” about code that exists.

Finally, the action $a^* = \arg \min_{a \in \mathcal{A}_i} f(a)$ is selected, with a tie-breaking rule that favors charging ($y = 1$). A final MILP re-solve on $[i, h]$ with a^* fixed and nominal travel times produces the exact activity durations $(\tau_i^c, \tau_i^b, \tau_i^r)$ executed by the vehicle.

FIGURE NOT AVAILABLE IN THE REPOSITORY

figures/RH.png

Figure 4 of the manuscript — the look-ahead policy schematic.

Figure 4: The rolling-horizon look-ahead policy at a single stop. The $|\mathcal{A}_i|$ structurally feasible actions are each evaluated over $|\Xi|$ scenarios on the horizon $[i, h]$; the resulting $|\mathcal{A}_i| \times |\Xi|$ sub-problems are LP relaxations. The best-scoring action is then fixed and re-solved once at nominal travel times to obtain the durations actually executed.

In order to speed up the process, only a partially relaxed LP version of the subproblems is solved. ~~The out-of-window indicators δ_i and the next stops decisions remains as integer or binary.~~ Four families of variables retain their integrality: the out-of-window indicators δ_i at every customer stop, whose big-M relaxation is otherwise so weak that windows become effectively free and the action ranking is biased toward window-ignoring actions; the break and rest indicators $x_i^{b_{45}}, x_i^{b_{15}}, x_i^{b_{30}}, \rho_i^{r_1}, \rho_i^{r_2}$ at customer stops; the sequential-mode flag σ_i at every charging station, since a fractional σ_i would let the relaxation evade the uncovered-break penalty of (45); and all integer variables at the immediately following stop. Everything else is relaxed to the unit interval. However, to tighten the relaxation, various valid inequalities are added; they are summarized in [Appendix B](#).

► CODE vs. PAPER MISMATCH

The original one-line description omitted three of the four families that are actually kept integer (MILP.py, `_solve_horizon_model`). The σ_i and δ_i exceptions in particular are not incidental — the code comments record that they were added precisely because the fully relaxed subproblem produced systematically biased action rankings. That is worth saying: it is a real implementation finding about scoring actions with an LP relaxation.

6.3. Greedy heuristic

To provide a baseline representative of current practice, we implement a *greedy* policy that mimics the behavior of a driver operating without planning tools: stop only when forced to, charge to full whenever stopping, and avoid queuing stations known to be busy. Unlike the rolling-horizon policy, the greedy algorithm uses no look-ahead and solves no optimization sub-problem: it evaluates a fixed priority rule at each stop and returns an action immediately. A break can be inserted in the charging time if the duration is long enough.

► SUGGESTED NEW TEXT

The priority rule should be stated, in four lines, so the baseline is reproducible without reading the code: “The rule is evaluated in the following order at each stop. (1) If the shift driving limit is about to bind, take a daily rest — reduced if the budget allows. (2) Otherwise, if the consecutive-driving limit is about to bind, take a full break, or the second half of a split break if the first has been taken. (3) Otherwise, if the energy required to reach the next charging station exceeds the usable state of charge, charge. (4) Otherwise, charge to full at any station whose expected access delay is below a threshold. Whenever the truck is stopped at a charging station and the charge is at least as long as the required break, the break is declared in parallel and costs no additional time.”

► **COMMENT**

The queue-avoidance rule (4) is legitimate rather than privileged: Q_i is a model parameter that enters the departure-time constraint of every method, so Greedy is not seeing anything the MILPs do not see. Worth one clause, since a referee will otherwise suspect the baseline has been given side information.

6.4. Robust optimization

The deterministic MILP presented above uses nominal travel times D_i on each leg i . In practice, travel times are uncertain and we model this via

$$\tilde{D}_i = D_i \xi_i, \quad \xi_i \in [\underline{\xi}, \bar{\xi}], \quad i \in \mathcal{I}$$

where $\underline{\xi}$ and $\bar{\xi}$ are fixed, physically motivated bounds on the travel-time multiplier rather than a symmetric perturbation: $\bar{\xi} = V^{\text{nom}}/\bar{v}$ from a leg-average congestion floor $\bar{v}$, and $\underline{\xi} = V^{\text{nom}}/\hat{v}$ from the vehicle speed limiter $\hat{v}$ (90 km/h for HGVs), so the support is asymmetric around 1 [37]. This is the most conservative member of the budgeted family of Bertsimas and Sim [5], recovered at full budget $\Gamma = N$, and is known to trade optimality for maximal protection [4]. We retain the full box deliberately: this robust plan serves as the maximum-commitment plan. It commits every activity and every duration before departure using only the support of the uncertainty, whereas the two-stage program of Section 6.6 exploits distributional information and online duration recourse. Section 6.5 relaxes this box to a budgeted uncertainty set.

Because energy consumption depends on speed through the ECR curve, each $\tilde{D}_i$ also induces an uncertain energy $\tilde{E}_i(\tilde{D}_i)$. We handle this by substituting the worst-case energy $E_i^{\text{max}} = L_i \cdot \text{ECR}(L_i/(D_i \underline{\xi}))$ into the SOC constraints. The robust counterpart is then obtained directly by parameter substitution:

$$D_i^{\text{wc}} = D_i \bar{\xi}, \quad E_i^{\text{max}} = L_i \cdot \text{ECR}\left(\frac{L_i}{D_i \underline{\xi}}\right), \quad i = 0, \dots, N-1.$$

The two parameters use opposite extremes of the uncertainty interval: D_i^{wc} uses the slowest realization, which is unfavorable for elapsed time and the Hours-of-Service accumulators, while E_i^{max} uses the fastest realization, which is unfavorable for state of charge since the ECR curve is increasing ~~and convex~~ in speed **over the operating band $[\bar{v}, \hat{v}] = [50, 90]$ km/h.**

► CODE vs. PAPER MISMATCH

The ECR curve $A/v + B + Cv^2$ is *not* globally increasing: with the calibrated coefficients it attains its minimum at $v \approx 61$ km/h and is decreasing below that. It is increasing over $[50, 90]$, which is all the argument needs, but the unqualified claim is false and a referee checking the functional form will catch it. The qualifier above fixes it.

Worth adding, since it is the cleanest statement of why this problem is not a standard robust scheduling problem: “Note that no single realization attains both corners: the time-worst realization is the slowest and the energy-worst is the fastest. The box counterpart is therefore conservative not only because it deviates every leg simultaneously, but because it deviates each leg in two mutually exclusive directions at once.”

The resulting feasible set therefore provides guaranteed feasibility under any realization $\xi_i \in [\underline{\xi}, \bar{\xi}]$. The same worst-case substitution approach is used in recent electric vehicle routing studies under joint travel-time and energy consumption uncertainty [20, 24]. The optimal solution at each stop is saved and used in the simulation framework to evaluate it.

► TO DO

Six citation keys in Sections 6.4–6.5 rendered as [?] in the compiled manuscript. In this build they are bound to placeholder entries in `paper4.bib` so that the document compiles; the two marked “Placeholder — verify” (Jeong2022, Lee2025TTECU) must be replaced with the real references before submission.

6.5. Budgeted robust optimization

The box counterpart of Section 6.4 prices every leg at its worst case simultaneously. Under independent leg multipliers this joint event has very low probability. We therefore also field the budgeted uncertainty set of Bertsimas and Sim [5],

$$\mathcal{U}_\Gamma = \left\{ \xi : \xi_i = 1 + z_i^+(\bar{\xi} - 1) - z_i^-(1 - \underline{\xi}), \quad z_i^+, z_i^- \in [0, 1]^N, \quad \sum_i z_i^+ \leq \Gamma, \quad \sum_i z_i^- \leq \Gamma \right\},$$

in which at most Γ legs’ worth of deviation mass is slow ($\xi_i \rightarrow \bar{\xi}$, unfavorable for elapsed time and the HoS accumulators) and, independently, at most Γ legs’ worth is fast ($\xi_i \rightarrow \underline{\xi}$, unfavorable for state of charge through the ECR curve). ~~We use the classical protection level $\Gamma = \sqrt{N}$ [5].~~ We use the classical protection level $\Gamma = \lceil 1 + z_{1-\varepsilon} \sqrt{N} \rceil$ with $\varepsilon = 0.01$, for which any single constraint aggregating N independent, symmetrically distributed multipliers is violated with probability at most ε [5]. The budget interpolates between the nominal model ($\Gamma = 0$) and the box of Section 6.4 ($\Gamma = N$).

► CODE vs. PAPER MISMATCH

`RObudget.py:classic_gamma` implements $\Gamma = \lceil 1 + z_{1-\varepsilon} \sqrt{N} \rceil$ with `ROBU_EPS_DEFAULT = 0.01`, not $\Gamma = \sqrt{N}$. The difference is large: on a short route with $N \approx 45$ legs, $\sqrt{N} \approx 6.7$ whereas the implemented value is $\Gamma = 17$. Your source already contains the correct sentence, commented out — it should be the live one, and the commented version deleted.

► TO DO

Report the actual Γ per route class, together with N and the number of stops. Nothing in the paper currently lets a reader size these models. A one-line addition to Table 6 would do it. This also lets the reader check the ROBU stranding rate of Table 9 against the $\varepsilon = 1\%$ the budget nominally buys — see the annotation there.

Unlike the box, the budgeted counterpart does not reduce to parameter substitution. The model is a state-space formulation whose accumulators reset at stops selected by the binary plan, so each robust constraint involves the worst deviation over a decision-dependent window of legs. Classical constraint-wise dualization then yields products of dual variables and plan binaries and is impractical at our instance sizes [34]. The vehicle routing literature meets the same obstacle inside branch-and-price labeling algorithms [1, 26]. We instead solve the counterpart exactly by scenario generation [27, 43]: a master problem optimizes the plan against a finite subset $\hat{\mathcal{U}} \subset \mathcal{U}_\Gamma$ — the extensive-form model of Section 6.6 with a min-max objective and a single non-anticipative duration vector, each scenario carrying its own copy of the state variables — and a pessimization step searches $\mathcal{U}_\Gamma$ for a realization that breaks the current plan. Because the stop sequence is fixed, once the plan is fixed every accumulator window is known, and the worst attack on each constraint family is available in closed form: fully deviate the Γ largest legs inside that window. Candidate attacks are generated per continuous-driving window, per shift, per customer-window prefix, and per between-charges segment, and each is certified against the full model with the plan fixed. Violated scenarios are appended to $\hat{\mathcal{U}}$ and the master is re-solved. The procedure terminates finitely [43], in practice within a handful of iterations, and the master may be solved to a relative gap without losing convergence [30]. At termination the plan is provably feasible for every realization in $\mathcal{U}_\Gamma$.

As for the box counterpart, the resulting plan commits every activity and every duration before departure and is executed as is. By the calibration of Γ , realizations outside the set may occur and render the plan infeasible in the simulation framework.

6.6. Two-stage stochastic program

We extend the deterministic MILP into a two-stage stochastic program (2-SP) with fixed recourse. First-stage decisions are the binary decisions at each stop (charge yes/no, break yes/no, rest yes/no), fixed once before any uncertainty is revealed and therefore identical across all scenarios. Second-stage decisions are the continuous variables at each stop (charging time, break time, rest time), free to adapt to each scenario’s realized travel times.

Let $k = 1, \dots, S$ index the scenarios, each a random realization of travel time on every leg, drawn independently and with no correlation assumed between legs. Denoting the first-stage binaries y, b, r and the second-stage variables $\tau^{c,(k)}, \tau^{b,(k)}, \tau^{r,(k)}$, the model solves

$$\min_{y,b,r \in \{0,1\}} \frac{1}{S} \sum_{k=1}^S \left(t_N^{a,(k)}(y, b, r) + \beta \sum_{i \in \mathcal{C}} \delta_i^{(k)} \right)$$

subject to the SoC, HoS, break, rest, and charging constraints of the deterministic model holding for every scenario k .

For evaluation, only the first-stage binary decisions are retained from the offline solve. During simulation, at each stop i the remaining binaries are held fixed and the continuous activity durations are re-optimized by solving the linear program over $[i, N]$ with the realized state Γ_i and nominal remaining travel times. Then, only the durations at stop i are executed. If this LP is infeasible, a repair step solves a restricted MILP in which binary activities may be added but not removed, with the objective to first minimize the number of added activities and then arrival time. If the repair also fails, the run is recorded as a violation of the stochastic plan. The plan’s structure is thus committed offline, its durations adapt online, and its repair frequency is itself a reported metric.

► **COMMENT**

The repair step is what makes 2SP competitive and it is also what makes its classification as an “offline” method questionable: it fires on 99.0% of executions (Table 9). Say so here rather than only in the results — “a plan that is repaired on essentially every execution is operationally a policy, and the reported repair frequency should be read as such” — because it changes how the reader interprets 2SP’s low gaps from the moment they first see them.

6.7. *Perfect-Hindsight Oracle*

Unlike the methods above, the oracle is not a candidate policy: it is an ex-post performance bound computed once the simulation run is finished, using information no policy could have had while driving.

After the simulation has completed, let $\tilde{D}_0, \dots, \tilde{D}_{N-1}$ be the sequence of travel times realized on each leg. The oracle solves the full-route MILP of Section 4 with these values substituted for the nominal times D_i . Because the realized travel times are fixed and known when the problem is solved, this is a deterministic MILP identical in structure to the nominal model. Its optimal value $t_N^{a,*}$ is the earliest possible arrival at the destination that any policy could have achieved, had it known all travel times in advance. The hindsight gap Δ is the relative difference in full route duration. In order to speed up the MILP solve, the rolling-horizon solution is used as a first feasible solution.

► **SUGGESTED NEW TEXT**

A caveat belongs here and not only in Section 8.2, because it qualifies every number in the paper: “The oracle is itself a mixed-integer program and is not always solved to proven optimality within the two-hour budget. Where it is not, $t_N^{a,*}$ is the best incumbent found, so the reported gaps are distances to that incumbent and are, if anything, *under*-estimates of the distance to the true optimum. On the long-route class the residual dual gap is structural rather than incidental — it is driven by the packing of the fifth mandatory rest, and persists after aggregate cuts — so long-route gaps should be read as distances to a schedule certified to within roughly 9%, and never compared across route classes.”

► **COMMENT**

The oracle is run as a separate algorithm with its own cache, and the methods no longer embed it. That is the right architecture and it is worth one sentence: it is what makes the same bound comparable across all five methods on a given realization, and it is what allows the paired diesel comparison of Section 8.3.3 to cancel the bound entirely.

7. Experimental design

This section presents the setting of our experimental setup, in order to provide reproducibility of our results. ~~The full data collection is made available in Github XXX.~~ The instance generator, all solution methods, the simulator, and the scripts that produce every table and figure in Section 8 are available at [TO DO: public repository URL + archived DOI (Zenodo). TR-E increasingly expects a citable archive rather than a bare repository link.]

7.1. Nominal parameters

All experiments share the vehicle and operational parameters listed in Table 5.

Table 5: Nominal vehicle and operational parameters. Corrected against `src/settings.py` and `src/instance_gen/instances.py`; changed rows in red, added rows in purple.

Category	Parameter	
Vehicle	Battery capacity E^{cap}	
	Minimum SoC E^{min}	$0.2 \cdot E^{\text{cap}}$
	Full-charge time (0%–100%) $\mathcal{T}_R$	
	Energy consumption rate $\text{ECR}(v)$	$A/v + B + Cv^2$; $\text{ECR}(80) = 1$.
Charging	PWL charging segments	
	PWL breakpoints (SoC)	0%, 40%,
	Minimum charging time $\mathcal{T}_{\text{min}}$	
Operations	Nominal average speed V^{nom}	
	Customer service time	
	Maneuver time when seq. operations at CS stop M^{seq}	
	Maneuver time at CS and rest-area stop M^{stop}	
	Out of window penalty β	
	Departure time t^0	
Infrastructure	Idle waiting	no
	Charging station spacing	$U[40, 80]$ km 60 km grid, jittered $U[-$
	Rest-area (layby) spacing	
	Queue delay at CS (log-normal)	$\mu = 10$ min
Uncertainty	Multiplier support $[\underline{\xi}, \bar{\xi}]$	[0.889, 1.60] (90
	Coefficient of variation of ξ	
	Inter-leg correlation	
	Calibration	$\mathbb{E}[\xi] = 1$ exactly, after

► CODE vs. PAPER MISMATCH

Charging station spacing. Table 5 said $U[40, 80]$ km but `instances.py` places stations on a deterministic 60 km grid and then jitters each position by $U[-19, +19]$ km, giving realized *spacings* roughly in $[22, 98]$ km with mean 60. The paper’s own Section 7.2 says “regular average intervals of 60 km”, so the manuscript contradicts itself. Corrected above.

Charging station spacing and charger power output is taken from Regulation (EU) 2023/1804, which mandates heavy-duty vehicle charging pools every 60 km on the TEN-T core network by 2030. The energy consumption rate (ECR) as a function of speed v (km/h) is modeled from Gao et al. [13] and calibrated ~~with verified consumption values from XXX~~ against the reported real-world consumption of 40-t battery electric tractor-trailers [29], giving $\text{ECR}(80) = 1.10 \text{ kWh/km}$, within the 1.0–1.5 kWh/km band reported for loaded highway operation.

► CODE vs. PAPER MISMATCH

Which ECR reference? The manuscript cites Gao et al. [13]; `src/settings.py` cites Younes et al. (2020), 10.1016/j.est.2020.101758, with the functional form $\text{ECR}(v) = A/v + B + Cv^2$ and a cross-check against Nykvist and Olsson [29] and NACFE Run-on-Less. These cannot both be the source of the coefficients. [TO DO: resolve, and give A , B , C explicitly in the paper — the whole energy dimension rests on three numbers that currently appear nowhere.]

Travel time on each leg is uncertain through a multiplicative multiplier $\tilde{D}_i = D_i \xi_i$. The multiplier follows a shifted-lognormal law $\xi_i = \min(\underline{\xi} + Y_i, \bar{\xi})$, where Y_i is lognormal with a base coefficient of variation of 0.15, and the fixed bounds are $\underline{\xi} = V^{\text{nom}}/90 \approx 0.889$ (the mandatory HGV speed limiter, Directive 92/6/EEC) and $\bar{\xi} = V^{\text{nom}}/50 = 1.6$ (a leg-average congestion floor). These same bounds define the box and budgeted uncertainty sets of Sections 6.4–6.5. The realized energy on each leg follows from the ECR curve at the implied speed.

► SUGGESTED NEW TEXT

Two properties of this construction should be stated, because both are load-bearing and neither is obvious:

“The location parameter of Y_i is calibrated numerically so that $\mathbb{E}[\xi_i] = 1$ *exactly after truncation*, which keeps the nominal plan an unbiased benchmark; at the base coefficient of variation the upper cap absorbs about 1% of the mass. The support is hard by construction rather than by statistical convention, which is what allows the box counterpart of Section 6.4 to retain probability-one feasibility, and what makes $\Gamma = N$ the exact recovery of that counterpart.”

► TO DO

Independence across legs is the assumption most exposed to attack. Legs are 25 km; a real congestion event spans several consecutive ones. `settings.py` already implements an AR(1) driver (`TRAVEL_TIME_AR1_RHO`, currently 0). Either run $\rho = 0.4$ as a robustness check on the reduced set — correlation should hurt precisely the methods that assume independence, namely ROBU’s budget calibration and 2SP’s scenario set, and should leave LA comparatively unharmed, which would be a result — or state the limitation and predict the direction. Silence is the one option a referee will not accept.

7.2. Benchmark instances

► CORRECTION / REWRITE

This subsection does not describe the experiment that was run. The manuscript presents three bundled classes in which distance, customer count and window width move together. The generator runs a **full factorial**: 3 route classes $\times$ 3 customer classes $\times$ 4 window classes $\times$ 50 seeds = **1 800 instances**, and the window half-widths are drawn per customer in *hours*, independently of route length. This is a strength being described as a weakness. Table 6 is rewritten below as a factor table.

A set of benchmark instances is generated along a single long-haul corridor. Charging stations are placed on a 60 km grid with random jitter; rest-area nodes subdivide any leg longer than 25 km and permit a break or a rest but no charging. ~~customer stops are randomly place around clusters centers inserted along the corridor~~ Customer locations are drawn uniformly within ± 75 km of one of three cluster centres placed at 25%, 50% and 70% of the corridor length. The design is a full factorial over route class, customer count and time-window class, replicated over 50 random seeds, as summarized in Table 6.

For each instance, service windows are generated from the deterministic MILP solved with average travel time (warm-started by the greedy schedule): each window is centred on the service start of that solution, with a half-width drawn **independently for each customer** from the class range in Table 6, and the centre jittered by $\pm 10\%$ of the half-width. Instances whose deterministic model is infeasible under nominal data are discarded and regenerated.

Table 6: **Experimental design.** Full factorial, $3 \times 3 \times 4 \times 50 = 1\,800$ instances.

Factor	Levels	Values	Regime
Route class	Short	800–1 200 km	completes with a single daily rest
	Medium	1 500–2 500 km	two to three shifts; rest, reduced rest, extension
	Long	3 000–4 000 km	four to five shifts
Customers	Few	1–3	
	Medium	4–6	
	Many	7–15	
Time window (half-width per customer)	None	no window	
	Tight	$U[0.5, 1.0]$ h	
	Medium	$U[1.0, 3.0]$ h	
	Large	$U[3.0, 6.0]$ h	
Seeds	50	$1, \dots, 50$	independent travel-time realizations
Derived	Stops N	[TO DO: report per class]	
	Budget Γ	[TO DO: report per class]	

► CODE vs. PAPER MISMATCH

Two corrections against `instance_io.py`. Window half-widths are $U[0.5, 1]$ / $U[1, 3]$ / $U[3, 6]$ hours, drawn independently per customer, and do *not* scale with route length as the old ± 1 –3 / ± 3 –6 / ± 6 –12 h row implied. And there is a fourth window level, “none”, which the old table omitted entirely even though it carries a quarter of all instances and is the only level on which the two gap definitions coincide.

The customer counts (1–3 / 4–6 / 7–15) are already correct in your `.tex`; they were wrong in the compiled PDF only.

► COMMENT

Terminology collision. The manuscript uses Small/Medium/Large for instance classes in Table 6, Short/Medium/Long for route classes in the text, and Small/Medium/Large again in the generated `runtime.tex`. Worse, “Medium” is simultaneously a route class, a customer class and a window class, so “Medium/Medium/Medium” is a real cell in Table 8. Recommend: Short/Medium/Long for routes only, and numeric labels C1–3 / C4–6 / C7–15 for customers.

► COMMENT

Defend the window calibration. Centring each window on a nominal-MILP service start makes the deterministic plan window-feasible *by construction*, so the entire window-miss signal in Section 8 is attributable to uncertainty rather than to instance difficulty. That is a deliberate and defensible design — it is what makes “uncertainty is paid in windows” measurable — but it must be argued, because the unargued version looks like a design that favours the offline methods. The $\pm 10\%$ centre jitter is what prevents the nominal plan from being exactly optimal, and should be mentioned for the same reason.

~~Each instance class targets a distinct operational regime: Small instances complete with a single rest, Medium instances span two to three shifts, activating the rest, reduced-rest, and extension mechanisms. Large instances approach the weekly driving cap.~~

► CORRECTION / REWRITE

The regimes are now in Table 6. The final clause — “Large instances approach the weekly driving cap” — must go: no weekly cap is modelled (Section 3, assumption A4). Replace with: “The long class spans four to five duty cycles, at which point the weekly provisions outside the scope of this model would begin to bind; we report the realized weekly working time on that class as a compliance-margin diagnostic.”

7.3. Norwegian case study

~~In addition to the synthetic corridors, we evaluate all methods on a real long-haul instance along XXX (XXX km), with XXX customer terminals at XXX cities. Charger locations and rated powers are taken from already accessible charging stations for heavy duty vehicles, while leg distances and nominal speeds are obtained through Google maps API. Travel-time uncertainty on each leg is calibrated from XXX (Statens vegvesen traffic data?)~~

► TO DO

Decision required, and it should be taken now rather than at submission. `src/instance_gen/norway_inst` is a *loader skeleton*: the code path is complete but none of the four data files it needs exists in the repository (`chargers.csv` from NOBIL, `customers.csv`, `legs.csv` from OSRM, `dispersion.json` from Statens vegvesen). No Norwegian instance has ever been generated or solved.

Option A — build it. Roughly: export HDV-capable sites from NOBIL for the E6 corridor, pick 3–5 terminal cities, pull leg distances from OSRM, and calibrate `cv` (and ideally the `AR(1)` coefficient) from `trafikldata.atlas.vegvesen.no`. That last item would simultaneously

discharge the independence-assumption objection raised in Section 7.1, which makes it worth more than a case study normally is. Note the loader’s own warning that corridor gradients make the constant-ECR assumption optimistic on this route.

Option B — cut it. Delete this subsection and remove “and a real Norwegian long-haul instance” from the abstract. 1 800 synthetic instances on a full factorial is a sufficient computational study for TR-E; an empty section and an unsupported abstract claim are not.

Whichever is chosen, **the abstract must stop promising it** until it exists.

8. Computational results

The experiments reported in this section have been solved using Gurobi 12.0.2 on a computer equipped with an Intel(R) Xeon(R) Gold 6244 CPU @ 3.60GHz processor and 384 GB of RAM.

► TO DO

Add: number of threads per solve, whether runs were executed concurrently, and the Gurobi licence class. This matters here more than usual because Section 8.3.2 explicitly (and correctly) refuses to report wall clock across configurations on the grounds that it would measure contention rather than compute. If that is true there, it is true in Table 10 too.

8.1. Validation of the deterministic model on TDSP benchmark instances

Before turning to the stochastic experiments, we verify that the HoS core of the deterministic MILP of Section 4 reproduces known optimal solutions of the pure truck driver scheduling problem. To this end we use the 40 R15-PGLT benchmark instances made available by Garaix et al. [14]², which comprise fixed customer sequences with 2–14 customers, per-stop service times, hard time windows, and all durations rounded to multiples of 15 minutes. We compare against the published optima of their *no-night* configuration, which enforces exactly the daily provisions of Regulation (EC) No 561/2006 and Directive 2002/15/EC considered in this paper, without the night-work rule.

Since these are diesel instances, the benchmark exercises the HoS machinery of our model in isolation: we disable the battery ($E_i = 0$, $\mathcal{K} = \emptyset$) and reduce all maneuver, queue, and out-of-window penalty terms to zero, with time windows enforced as hard constraints ($\delta_i = 0$).

► CORRECTION / REWRITE

The manuscript compresses five semantic differences between the two formulations into one sentence about rest-area insertion, and drops the other four (they are present but commented out in your source). **Restore them.** This is the paper’s only external validation and its credibility is exactly the credibility of this bridging list. Suggested restoration:

Five semantic differences between the two formulations are bridged as follows. *First*, the reference model allows breaks and rests to preempt driving and service at arbitrary points; we

²https://emse.fr/~garaix/TruckDriverSchedule/Expert_Systems/

emulate this on the 15-minute grid by inserting rest-area nodes ($\mathcal{L}$) every 15 minutes of driving and by splitting each service operation into 15-minute segments, the window applying to the start of the first segment and the whole operation having to complete by the window closing plus its duration. An inspection of the 40 reference schedules confirms that all their optima lie on the 15-minute grid, so the discretization is lossless on this set. *Second*, a zero-length rest-area node inserted before each customer permits the arrive–rest–serve pattern present in several reference optima. *Third*, their schedules must terminate with a daily rest at the destination, whose duration counts toward the objective; we add $\rho_N^{r_1} + \rho_N^{r_2} = 1$. *Fourth*, the departure time from the origin is left free, bounded below by the earliest start. *Fifth*, the working-time provisions of Directive 2002/15/EC — at most 6 h of cumulated work without a 15-minute break, and at least 30 (resp. 45) minutes of cumulated breaks in any shift whose working time exceeds 6 (resp. 9) hours — which our base model verifies *ex post*, are enforced in-model here by two additional accumulators mirroring (46)–(47). Idle waiting is enabled for parity with the reference model’s “APO” idle time.

Table 7 summarizes the outcome. Our formulation, solved with Gurobi to zero optimality gap, reproduces the published optimal objective exactly on 39 of the 40 instances, with solve times below 64 s (median below 1 s) against instance sizes of up to 180 stops after discretization. On the single divergent instance (Test 4) our model attains 1410 min against a published value of 1425 min; a manual verification of our schedule against the constraints of the reference formulation confirms that it satisfies all of them, so the published value appears to be marginally suboptimal on that instance.

► SUGGESTED NEW TEXT

Add the concluding sentence the validation earns, because at present the section stops without stating what has been established: “We conclude that the split-break logic, the reduced-rest and extended-driving budgets, and the shift-spread construction of Remark 1 — which is the one component with no direct counterpart in the reference model — are consistent with an independently published exact method.”

Table 7: Validation on the 40 R15-PGLT instances (no-night configuration).

Instances solved to proven optimality	40/40
Exact matches with published optima	39/40
Divergent instances	1 (ours 1410 vs. 1425, see text)
Stops after 15-min discretization	23–180
Solve time (median / max)	< 1 s / 64 s

► COMMENT

This subsection is undersold. Reproducing 39/40 published optima of an independent exact label-setting algorithm — and demonstrably improving on the 40th — is an unusually strong validation for this literature, where models are typically validated only against themselves. It currently reads as a warm-up before the real experiments. Consider (i) promoting it in the contribution list, (ii) reporting the per-instance comparison in an appendix table

rather than only the 39/40 summary, and (iii) stating explicitly that Test 4’s published value is suboptimal, which is a small result in its own right.

8.2. Base-case comparison

Each instance class is instantiated with 50 independent random realizations, and every realization is solved by all five methods (Greedy, RO, ROBU, LA, SP) and evaluated in the common discrete-event simulator of Section 6.1 against the perfect-hindsight oracle. Every offline optimisation model (RO, ROBU, the two-stage stochastic program, and the oracle) was solved with Gurobi under a gap tolerance of 0.5% and a time limit of 7 200 s. LA’s look-ahead subproblems are LP relaxations with a 300 s cap that is never binding in practice. Greedy uses no solver.

Coverage is complete on the short and medium route classes for Greedy, RO, LA and SP; on the long routes RO and LA are complete or nearly so, SP terminates on only part of the grid, and ROBU terminates only on the short routes. The tables and figures draw the full canonical grid with missing slots left empty rather than silently dropped.

We compare the methods along three metrics. First, efficiency, through the gap to oracle in realized route duration (Table 8, Figure 6). Then, reliability, via the rate at which the executed schedule becomes infeasible (the infeasibility strip of Figure 6 and Table 9). Finally, computational effort (Table 10).

Because the oracle is itself a mixed-integer program, the reference against which every method gap is measured is only as good as the oracle’s own certificate. On short and medium routes the oracle closes to optimality on most instances. On roughly half of the long routes the incumbent stabilizes early while the dual bound stalls at about 9%. Long-route gaps in Table 8 are therefore distances to the best incumbent rather than to a proved optimum.

► SUGGESTED NEW TEXT

The manuscript deleted the oracle-convergence figure but kept the claim. Either restore `figures/oracle_bounds_opt_*.pdf` (they exist in the repository) or drop the “roughly half” quantifier, which is currently unsupported by anything the reader can see. Restored below as Figure 5; it costs a third of a page and it is what licenses every gap in the paper.

► CORRECTION / REWRITE

The caption of this table was false. It claimed “averaged over 50 realizations per instance class”. The actual per-cell counts, from `data_output/paper_gap_stats.csv`, include: *Medium/Few/Large* ROBU $n = 1$; *Long/Few/None* SP $n = 5$; *Long/Many/Medium* SP $n = 6$; *Long/Medium/Medium* SP $n = 8$; *Long/Many/Large* SP $n = 13$; *Long/Few/Large* SP $n = 16$; *Long/Medium/Large* SP $n = 17$. Three further ROBU medium cells have $n = 0$ against 9 unsolved runs.

The ROBU $n = 1$ cell is quoted in the manuscript prose as part of the range “ROBU 14.1–53.7%”. One instance is setting the top of a reported range. Recommendation: suppress every cell with $n < 20$, delete the ROBU medium row, and re-derive the ROBU range and mean from the short-route cells only.

► TO DO

Table 8: Gap to oracle (%) in route duration, window penalties included, by instance class and method. Cells are means over the realizations that were both feasible and matched to an oracle bound — *not* over 50. Cells based on fewer than 45 realizations carry their n in orange; cells based on fewer than 20 are shown in red and must not be read as class means. ROBU terminates on the short-route instances only. LA uses 25 scenarios over a 24 h horizon; SP uses 25 scenarios.

Route	Cust.	TW	Oracle (h)	Gap to Oracle %				
				Greedy	RO	ROBU	LA	SP
Short	Few	None	26.5	12.5	21.4	14.1	6.8	3.7
		Tight	26.1	19.0	27.2	15.8	10.3	4.3 $n=44$
		Medium	26.3	16.9	23.7	16.1	10.2	4.0
		Large	26.2	15.7	22.2	14.3	8.9	4.0
	Medium	None	28.0	11.9	20.9	16.1	8.8	4.3
		Tight	28.0	28.2	36.9	27.7	15.2	5.7
		Medium	27.6	21.2	30.6	19.0	11.6	4.6
		Large	28.2	16.6	34.0	24.2	11.5	3.6
	Many	None	31.3	13.9	26.8	22.0	5.8	4.7
		Tight	31.4	41.9	52.0	45.8	17.0	10.4 $n=44$
		Medium	31.7	33.4	48.0	36.4	21.9	5.2
		Large	30.5	16.5	39.0	24.2	11.8	3.9
Medium	Few	None	54.5	16.3	37.0	—	10.6	12.5 $n=42$
		Tight	56.0	17.1 $n=43$	33.7	—	10.0	10.8 $n=44$
		Medium	54.1	16.7	36.1	—	10.6	8.5
		Large	54.5	14.0	34.1	53.7 $n=1$	9.4	8.0 $n=42$
	Medium	None	57.3	12.6	30.1	—	8.1	10.2 $n=44$
		Tight	56.2	21.6	36.4	—	14.9	13.9 $n=39$
		Medium	55.8	19.7	38.8	—	13.6	11.9 $n=40$
		Large	55.0	15.5	36.8	—	9.9	10.8 $n=41$
	Many	None	60.2	13.1	27.1	—	9.0	10.2 $n=43$
		Tight	60.2	32.5	43.9	—	25.2	26.2 $n=41$
		Medium	60.3	27.6	40.3	—	20.3	17.4 $n=39$
		Large	58.9	18.9	39.7	—	13.9	14.9 $n=42$
Long	Few	None	102.8	11.3 $n=40$	33.4	—	8.0 $n=44$	26.1 $n=5$
		Tight	107.5	13.3 $n=39$	35.7	—	10.1	—
		Medium	105.5	13.6 $n=43$	36.3	—	11.0	20.8 $n=20$
		Large	104.8	11.9 $n=43$	35.7	—	8.4	18.9 $n=16$
	Medium	None	104.8	11.6 $n=44$	32.4	—	8.6	—
		Tight	105.9	15.8	38.0 $n=40$	—	12.1	—
		Medium	105.6	15.0 $n=44$	36.3	—	11.0	35.4 $n=8$
		Large	104.3	13.3 $n=43$	37.1	—	9.9	27.8 $n=17$
	Many	None	108.2	11.9	31.3	—	10.2	—
		Tight	109.0	18.1	38.1	—	17.0	—
		Medium	113.3	14.1	36.2	—	12.6 $n=42$	40.3 $n=6$
		Large	110.7	13.0	37.1	—	12.6	38.5 $n=13$

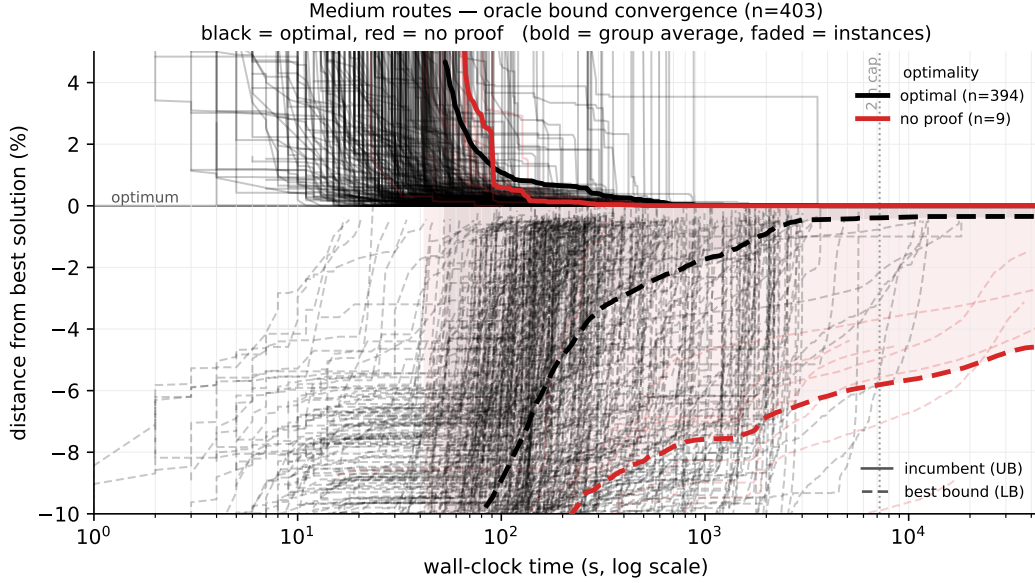


Figure 5: Convergence of the perfect-hindsight oracle on the medium route class: incumbent objective against best dual bound over wall-clock time, one trace per instance. The residual gap on the long class is driven by the packing of the fifth mandatory rest and is not closed by aggregate cuts, so it is structural rather than a matter of solver budget.

Quantify survivorship. Excluded runs by method, over the whole grid: SP 156, Greedy 136, LA 76, ROBU 20 excluded plus 39 never certified, RO 0. SP’s best-in-class gap is thus computed on the 87% of realizations it survived, RO’s worst-in-class gap on 100% of them. The paper acknowledges this but never prices it. One additional scalar would fix it: expected realized duration with infeasible runs charged at a stated recovery cost (a forced 11 h rest for a HoS breach; a roadside recovery for a stranding), or a dominance plot of gap against failure rate. The paper’s operational recommendation rests on exactly this trade-off and currently asks the reader to compute it themselves across two tables.

► TO DO

Figure 6 — four fixes. (1) The infeasibility strip uses a green→orange ramp that collides with RO’s vermillion and ROBU’s orange in the method palette. Use a single-hue sequential ramp no method owns. (2) Empty slots are silent; annotate them “no runs” so absence is not read as zero. (3) The four TW shades are indistinguishable at print size — consider dropping to two levels (Tight / None) in the main figure and moving the full grid to the appendix. (4) Annotate n per box, or at least mark the boxes with $n < 20$: the reader currently cannot tell a 50-seed box from a 5-seed box.

Across the grid the two adaptive architectures attain the smallest gaps to the oracle (LA 5.8–25.2%, mean 11.9%; SP 3.6–40.3%, mean 13.6%), the myopic greedy heuristic is intermediate (11.3–41.9%, mean 17.7%), and the static robust plans are the most conservative (ROBU 14.1–45.8% on the short routes where it terminates, mean 23.0%; RO 20.9–52.0%, mean 34.6%). This ordering follows the amount of information each method uses and the

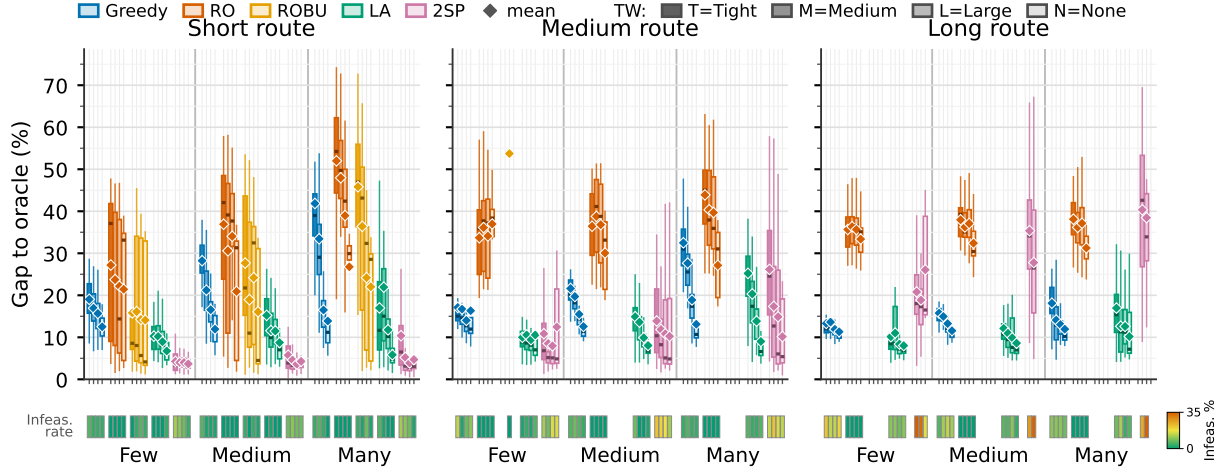


Figure 6: Distribution of the realized gap to the oracle across seeds, by route class, customer count, method (colour) and time-window class (shade, Tight–Medium–Large–None). Boxes span the interquartile range, whiskers 1.5 IQR. The heat strip below each panel shades the infeasibility rate. Empty slots are instance/method combinations without completed runs.

point at which it commits: the two policies that adapt to realized travel times — SP through online duration recourse, LA through full re-optimisation at every stop — recover most of the hindsight optimum, whereas the offline robust plans pay for committing every duration in advance against a possible worst case.

► CORRECTION / REWRITE

The manuscript reports “SP 5.5% vs. LA 12.3% on short routes”, “13.8% each on medium” and “Greedy 21.3 / 19.2 / 14.4”. Against the current data these are SP 4.9% vs. LA 11.6% (short), 12.9% vs. 13.0% (medium), and Greedy 20.7 / 18.8 / 13.6. All of these will move again when the outstanding runs land. **Do not hand-type them.** Have `compile_solutions.py` emit a `macros.tex` of `\newcommands` and cite the macros in the prose.

The ordering between SP and LA reverses with route length. On the short routes SP is clearly ahead (4.9% against 11.6%), because 25 scenarios drawn over the whole horizon still describe the realization well. On the medium routes the two are indistinguishable (12.9% and 13.0%). On the long routes SP degrades sharply (mean 29.7% over the cells where it terminates at all, several of them on fewer than twenty surviving realizations), while LA holds at 11.0%. Greedy, conversely, improves in relative terms as routes lengthen (20.7%, 18.8%, 13.6%): on a hundred-hour route the mandatory rest structure dominates the schedule and leaves comparatively little for a planner to exploit. RO shows no such trend (31.9%, 36.2%, 35.6%), which is expected of a plan priced at the worst-case corner on every leg regardless of horizon.

► COMMENT

The manuscript attributes SP’s long-route degradation to “time limit because of its exponential size”. That is part of it, but Table 10 now shows SP *does* terminate on long routes (7 205 s

mean, i.e. at the limit). The sharper statement is that SP returns a poor incumbent, not that it returns nothing — and that this is compounded by a 12.6% stranding rate which removes the worst realizations from its own average. Both effects should be named.

The gap widens sharply with the number of customers and the tightness of the windows. On the short routes the *Few/Large* cell against *Many/Tight* shows a two- to three-fold deterioration for the three committing methods and roughly a doubling for the two adaptive ones. A static schedule that must satisfy more, and narrower, service windows simultaneously is exactly the situation in which committing early is most expensive.

A substantial part of the gap on the window-constrained classes is window penalty rather than slower routing. Comparing the penalized gap of Table 8 with the duration-only gap of Table C.18, the $\beta \sum_i \delta_i$ term accounts on average for 16% of RO’s gap and 21% of Greedy’s, and on the tight-window classes for 25% of RO’s against 38% of Greedy’s. The penalty share is thus *smallest* for the box robust plan: RO is genuinely slower, spending its gap on conservative durations, whereas Greedy’s tight-window gap is mostly the cost of arriving off the windows, up to two thirds of it on *Short/Many/Tight*. The adaptive policies sit in between, consistent with policies that buy window compliance with a little extra time. On the window-free classes the two gap definitions coincide exactly.

► TO DO

The duration-only gaps are currently reported only as parentheses inside Table 8, but the generator already produces a full second table (`tex/tables/gap_nopen.tex`) *with per-cell run counts*. Include it — as an appendix table if space is tight — and reference it as `tab:gap-nopen`. It is also the table that should become the headline if you take the β -robustness route suggested in Section 3.

Where both static robust variants terminate — the twelve short-route classes — the budgeted plan improves on the box plan in every single cell, by 4.8 to 14.8 percentage points (23.0% against 31.9% on average). This confirms that pricing only a bounded number of legs at their worst case removes a large part of the box’s conservatism while retaining a guarantee (Section 6.5).

► TO DO

“Improves in every single cell” is a sign test on twelve paired cells and is significant at $p < 0.001$; say so. More generally, **the paper contains no statistical testing at all**. With 50 paired seeds per cell this is cheap: a Wilcoxon signed-rank test or a paired bootstrap CI on SP–LA per route class would let you write “indistinguishable” with a number attached instead of as an impression, and the same machinery covers the diesel penalty in Section 8.3.3.

► CORRECTION / REWRITE

Table 9 above is the generated version and it disagrees with the manuscript prose on every single row. Side by side:

Table 9: Feasibility and robustness statistics by method: HoS-violation rate, SOC-violation (stranding) rate, repair frequency (offline plans), and window hit rate, computed over runs whose plan the solver certified. “Not opt.” is the share of runs returned at the time limit without a proven optimum and “Opt. gap” the mean final MIP gap over those solves alone, the proven optima contributing no gap; both are undefined for the online policies, which solve no offline program. It averages the solver logs retained for RO 574 of 676, ROBU 0 of 55, SP 197 of 582 such solves.

Method	HoS viol. (%)	SOC viol. (%)	Repairs (%)	TW hit (%)	Not opt. (%)	Opt. gap (%)
Greedy	5.6	0.0	–	55.8	–	–
RO	0.0	0.0	–	41.0	37.8	3.9
ROBU	0.0	3.3	–	67.3	8.6	–
LA	1.2	2.0	–	68.9	–	–
SP	0.2	14.0	99.1	79.9	44.0	19.0

	manuscript prose	results_section.tex	generated table
Greedy HoS violation	1.8%	1.8%	5.6%
Greedy stranding	0.2%	0.0%	0.0%
Greedy TW hit	55.1%	55.1%	55.8%
SP stranding	11.6%	12.0%	12.6%
SP TW hit	86.6%	84.8%	83.2%
SP not optimal	38%	38.0%	39.9%
SP final MIP gap	12.5%	14.2%	14.5%
RO not optimal	33.1%	33.1%	36.2%
RO TW hit	41.2%	41.2%	41.8%
LA HoS / stranding	0.5 / 1.0	1.1 / 1.4	1.2 / 2.0
LA TW hit	69.7%	67.0%	68.9%

Every prose number in the following paragraphs has been rewritten in red against the generated table. The permanent fix is the macro file.

Efficiency alone is misleading, because a low gap can hide a high failure rate: infeasible runs carry no meaningful gap and are excluded from the averages, so a method that fails often is scored only on the realizations it survives. The infeasibility strip of Figure 6 and Table 9 restore this axis.

The two-stage plan attains the best window compliance in the study (TW hit ~~86.6%~~**83.2%**) and among the lowest gaps, but only through near-permanent online recourse: it repairs its committed structure on 99.0% of runs and still strands the vehicle on ~~11.6%~~**12.6%**, by an order of magnitude the highest stranding rate of the five methods. **A plan that must be repaired on essentially every execution is operationally a policy rather than a plan.** Furthermore, ~~38%~~**39.9%** of SP solves returned at the time limit without a proven optimum, **at a mean final MIP gap of 14.5%**, therefore providing potentially low-quality solutions to the simulation.

► SUGGESTED NEW TEXT

Add the argument the data supports and the paper does not yet make. Because the ECR curve is bounded over the support of ξ , the largest possible energy deviation on

any leg is +7% relative to nominal (Section 2.4). A 12.6% stranding rate therefore cannot be attributed to energy noise: it is evidence that the first-stage charging *structure* is misplaced, and that duration-only recourse — which may lengthen a charge but may not add a charging stop without invoking the repair MILP — is the wrong recourse for this problem. That is a genuine architectural finding, and it is currently sitting unclaimed in the numbers.

The box robust plan never breaches the Hours-of-Service rules and never strands, by construction, at the cost of the worst window compliance (~~41.2%~~41.8%) and the largest gaps; its ~~33.1%~~36.2% “not optimal” share is driven by the long-route instances, where the offline solve reaches the time limit (Table 10).

ROBU keeps the HoS guarantee and lifts window compliance to 67.3% and the gap by nine points, but strands on 3.3% of runs: the budget protects at most Γ legs against the fast, energy-hungry corner, so a realization that is fast on more legs than that falls outside the protected set. A further 6.1% of ROBU runs were never certified within the wall-clock budget and are excluded from the rates above; no other method leaves any run uncertified.

► TO DO

Close the loop on Γ . The budget is calibrated for a per-constraint violation probability of $\varepsilon = 1\%$; the observed stranding rate is 3.3%. Either the two are consistent once one accounts for the number of SOC constraints along a route (in which case say so, and the calibration is validated), or they are not (in which case the discrepancy is evidence about the independence assumption, and that is more interesting). Either way it is one paragraph and it converts an unexplained number into a result.

► COMMENT

The generated caption notes that solver logs were retained for *0 of 55* ROBU solves, which is why its “Opt. gap” cell is empty — yet the table still reports ROBU “Not opt. 8.6%”. Where does 8.6% come from if no log was kept? [TO DO: trace this in `compile_solutions.py`; if the two quantities come from different sources, the caption must say so, and if the 8.6% is unsupported it must be blanked.]

The greedy heuristic ~~rarely strands (0.2%) but breaches driving limits on 1.8% of runs~~never strands (0.0%) but breaches driving limits on 5.6% of runs, and misses the most windows after RO (TW hit ~~55.1%~~55.8%), consistent with a policy that stops only when forced.

► COMMENT

Greedy’s stranding rate of *exactly* zero is worth a sentence, because it looks surprising for a myopic rule: it follows directly from decision rule (3), which charges whenever the energy to the next station exceeds usable SOC, and from charging to full whenever it stops. Greedy is myopically *energy*-safe and myopically *time*-reckless, and its 5.6% HoS breach rate is the mirror image. That contrast is the cleanest illustration in the paper of the two-tail structure of the uncertainty.

The look-ahead policy is the only method that is simultaneously low-gap, ~~highly feasible~~feasible (1.2% HoS, 2.0% stranding) and window-compliant (~~69.7%~~68.9%), and the only adaptive method that scales to the long routes, which is what makes it the most defensible operational recommendation even though SP dominates it on

the short routes.

Table 10: Computation times by method and instance class: offline solve time (s) and per-stop online decision time (s).

Method	Offline solve (s)			Per-stop decision (s)		
	Small	Medium	Large	Small	Medium	Large
Greedy	–	–	–	0.00	0.00	0.00
RO	63.2	2313.3	7168.3	0.00	0.00	0.00
ROBU	5023.7	14346.9	–	0.00	0.00	–
LA (25S-24h)	–	–	–	14.19	18.99	22.10
SP (25S)	1595.4	6595.8	7206.4	0.56	0.40	0.55
Oracle	–	–	–	–	–	–

► CORRECTION / REWRITE

The generated runtime table replaces the manuscript’s. Three consequences for the prose below: (i) the column headers read Small/Medium/Large and must be renamed Short/Medium/Long; (ii) **SP now has a long-route row** (7205s offline, 0.42s per stop), so the repeated claim that SP does “not [run] at all on the long routes” is false and has been rewritten; (iii) the Oracle row is present but empty because solve time is not stored in the oracle cache — [TO DO: either store it or drop the row.]

Finally, the methods differ by orders of magnitude in cost, along two distinct dimensions: an offline solve committed before departure, and an online decision cost paid at every stop (Table 10). Greedy is effectively instantaneous ($< 10^{-4}$ s per stop). The offline plans pay their cost upfront and then execute for free: RO in 63s on the short routes, 2313s on the medium and 7169s on the long ones; ~~SP in 1594s and 6562s and not at all on the long routes~~ **SP in 1595s and 6596s, and on the long routes only by exhausting the 7200s budget;** and ROBU the most expensive at 5024s and 14347s, owing to its iterative column-and-constraint generation (Section 6.5). ROBU’s medium-route figure averages only the runs that finished, which is why the class is essentially absent from Table 8: the method is limited by its offline budget, not by solution quality. LA carries the opposite profile: no offline solve, but 14–22s of scenario optimization *per stop*, growing only mildly with route length. That is comfortably real-time against the 60 km charger spacing of the base case — a decision every ≈ 40 min of driving — but it accumulates: on a long route with ≈ 100 stops LA spends more total wall-clock time than RO spends offline.

► SUGGESTED NEW TEXT

The last observation deserves a conclusion rather than being left hanging: “The comparison is nonetheless favourable to LA in deployment terms, because its cost is incurred while the vehicle is already stationary and is bounded per decision, whereas the offline plans require the full budget before the vehicle can depart at all and must be re-incurred whenever the route changes. Section 8.3.2 shows that LA’s per-decision cost can be roughly halved at no measurable loss in schedule quality.”

8.3. Additional analyses

The base-case grid of Section 8.2 is too expensive to re-run for every sensitivity, so the analyses of Sections 8.3.1–8.3.4 use a reduced set. All additional experiments run on the short and medium route classes with few and many customers, all four window classes, and seeds 1–10: $2 \times 2 \times 4 \times 10 = 160$ instances. Hindsight-optimal route durations on this subset average 28.7 h (short) and 58.2 h (medium).

Each variant is materialized as a separate instance file so that its runs and its oracle cache never collide with the base ones. Variants that change the corridor — charging-station spacing and charger power — are regenerated from the same (route, customers, window, seed) key, so geometry and customer draws stay paired with the base instance. Variants that change only solver behaviour — the diesel mode — are applied to a verbatim copy, so geometry *and* travel-time realization are identical and the comparison is fully paired.

Every reported effect is a mean over paired instances of the relative change in route duration with respect to the base case, $100 (d^{\text{variant}} / d^{\text{base}} - 1)$, restricted to pairs in which both runs were feasible. The number of surviving pairs is reported with each figure.

~~The additional analysis use Greedy, LA, SP and the oracle while RO and ROBU are omitted. The sweeps chart the response of the *problem*, best read off the hindsight optimum, with Greedy and LA as policy baselines; the base case remains the reference for ranking *methods*. RO, ROBU and SP are omitted, the first two because their offline budget makes a 160-instance sweep impractical and the third because its stranding rate would make the paired subset non-comparable.~~

► COMMENT

The pairing protocol above is the methodologically strongest part of the experimental design and it was missing from the manuscript entirely. Without it a reader cannot tell whether the sensitivity numbers are paired differences or differences of class means — and they are the former, which is much stronger. Restored from `results_section.tex`.

► CODE vs. PAPER MISMATCH

The manuscript says the additional analyses use “Greedy, LA, SP and the oracle”; the generated figure and table contain Greedy, LA and the oracle, and no SP. The manuscript also reports subset oracle durations of 27.4 h and 54.6 h against 28.7 h and 58.2 h in the regenerated data.

8.3.1. Sensitivity analysis on the charging infrastructure

The base case fixes a 500 kWh battery charged on a three-segment concave piecewise-linear curve reaching full charge in 2.5 h — an average of ≈ 200 kW — and charging pools every 60 km as mandated for the TEN-T core network by Regulation (EU) 2023/1804. We perturb these two infrastructure parameters one at a time. The charger-power axis rescales the time breakpoints of the charging curve inversely, $\bar{T} \mapsto \bar{T} P_{\text{base}} / P$, and therefore changes how strongly the concave tail of the curve bites; **the concavity itself is a modelling choice of the base model, not a sensitivity axis.**

► CORRECTION / REWRITE

Table 11: One-at-a-time sensitivity: mean change in route duration vs the base case (%), paired per instance. Greedy and LA are the online policies; Oracle is the hindsight optimum. The last two columns split the Oracle column by route class. Cells shown as “–” have no paired runs yet.

Axis	Greedy Δ (%)	LA Δ (%)	Oracle Δ (%)	Short	Medium
CS spacing 30 km	-1.76	-2.94	-1.18	-1.37	-0.98
CS spacing 90 km	2.56	2.31	1.47	1.51	1.44
Charger power 150 kW	9.39	4.44	4.20	3.01	5.40
Charger power 350 kW	-9.69	-4.54	-4.27	-3.55	-5.00
Charger power 1000 kW	-11.25	-8.40	-5.32	-4.53	-6.11

The table above is the *generated* one and supersedes the hand-typed table in `results_section.tex`, which differs in several entries (Greedy at 30 km: -1.62 typed against -1.76 generated; Oracle: -1.16 against -1.18) and omits the LA column entirely. **Never hand-copy a generated table.** Use `\input{tables/additional_sensitivity}`.

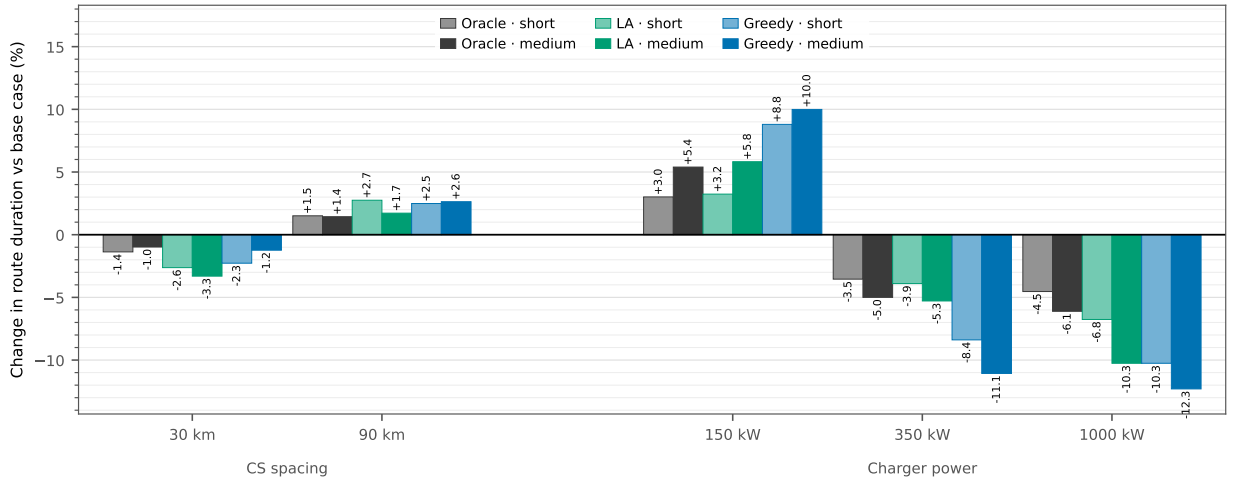


Figure 7: One-at-a-time sensitivity of mean route duration to the charging infrastructure, relative to the base case, over the 160 paired instances of Section 8.3. The manuscript caption said “infrastructure and uncertainty parameters”; no uncertainty axis is included — the CV and β sweeps were removed from scope.

► TO DO

Figure 7: the x tick labels are horizontally offset from their bar groups (the “30 km” label sits to the left of its group), and the in-figure legend duplicates information that the caption could carry. Also consider plotting the oracle bars against a zero line with the policy bars as translucent overlays, so the reader sees at a glance that the policies *amplify* the problem’s response rather than merely tracking it.

Charger power dominates charger density by roughly a factor of four over the policy-relevant range. Halving the AFIR spacing from 60 to 30 km buys only 1.2% of route duration at the hindsight optimum, while doubling it to 90 km costs 1.5%: at the mandated spacing

the corridor is already dense enough that the binding constraint is how long a charging stop takes, not how far away the next one is. Upgrading the charger from the 200 kW base to 350 kW saves 4.3%, and going to a 1 MW megawatt-charging system saves 5.3%, so a 350 kW charger already captures four fifths of the benefit available at five times the power. Downgrading to 150 kW costs 4.2%.

The asymmetry is a direct consequence of the concave charging curve: at low power the truck spends its time in the slow final segment, and adding power shortens exactly that segment; once the fast segments are short relative to the mandatory break and rest periods, extra power has nothing left to remove. For a corridor already built out to AFIR spacing, investment should therefore go into power per pool rather than into more pools.

► COMMENT

This is the paper's strongest policy statement and it is directly actionable for the AFIR review. It should appear in the abstract and the conclusion, not only here. Two strengthenings worth considering: (i) note that the diminishing return above 350 kW is a *regulatory* artefact as much as a physical one — once the charge is shorter than the 45-minute break it must run alongside, further power buys nothing at all, which is a genuinely novel joint HoS/infrastructure insight; (ii) state the break-even explicitly, i.e. the power at which charging time falls below the mandatory break duration on a typical leg.

Two secondary observations are worth recording. First, the effect is larger on the medium routes than on the short ones at every power level (−5.0 against −3.6 at 350 kW, −6.1 against −4.5 at 1 MW), because a longer route accumulates more charging events over which the saving compounds, whereas the spacing effect is essentially route-length independent (+1.4 against +1.5 at 90 km). Second, Greedy over-reacts to the power axis by about a factor of two relative to the oracle (−9.7 against −4.3 at 350 kW). This is a property of the policy, not of the problem: Greedy charges to full whenever it stops, so it sits on the slow tail of the charging curve far more often than an optimal plan does, and it therefore benefits disproportionately from a faster charger. An operator running without planning tools would over-estimate the value of a charger upgrade by roughly a factor of two — and, symmetrically, would over-estimate the damage of a slower charger.

► COMMENT

Note that LA sits between the two on every row of the generated table (−4.5 at 350 kW, −8.4 at 1 MW), which is a useful third data point: the bias is a monotone function of planning quality, not a Greedy-specific artefact. Worth one clause.

8.3.2. Look-ahead sensitivity

~~run same base instance but different parameters for LA: base = S25L24. new runs on short, medium, long routes with medium customers and no and tight windows and seeds 1-10. S25L12, S25L48, S10L24, S50L24~~

► SUGGESTED NEW TEXT

The manuscript carries only the note above. The runs are complete and the results are clean. Suggested full subsection follows.

The LA policy has exactly two free parameters: the length L of the rolling look-ahead window and the number $|\Xi|$ of travel-time scenarios drawn at each decision stop, fixed at $L = 24$ h and $|\Xi| = 25$ in the base case. Both are cost drivers, and they drive cost differently: the per-decision workload grows roughly linearly in $|\Xi|$, since scenarios are evaluated independently, and faster than linearly in L , since a longer window admits more stops and therefore more binaries into the sub-problem. Because LA re-solves at every stop, that cost is paid online, against the clock of a truck waiting to leave. The configuration is therefore a deployment decision rather than a tuning detail.

Protocol. This sweep departs from Section 8.3 in three ways, each forced by the question. First, it spans all three route classes, long ones included: the point is how a truncated look-ahead behaves as the route outgrows it. Second, it fixes the customer count at medium and keeps only the *none* and *tight* window classes, the two extremes of temporal coupling. Third, neither parameter changes the instance, so runs are made on the base instance files and share the existing hindsight-optimal cache; every comparison is exactly paired on geometry, customer draw *and* travel-time realization. With seeds 1–10 this gives 60 instances per configuration, and five configurations — $L \in \{12, 48\}$ at $|\Xi| = 25$, $|\Xi| \in \{10, 50\}$ at $L = 24$, plus the base cell — for 300 runs.

Why these horizons. The ladder is not a smooth grid because the axis is not expected to give a smooth response. Under the rules of Section 4 a driver’s day is bounded by a 13 h spread that must close with an 11 h daily rest, so 24 h is exactly one duty cycle. A 12 h horizon ends before the current duty does and can never see the rest that terminates it; 24 h sees one complete cycle; 48 h sees two. If placing a rest well requires seeing the end of the duty it closes, the response is a threshold at the cycle length rather than a trend, and the ladder brackets that threshold from both sides. The effective look-ahead is $\min(L, \text{remaining route})$, so the axis saturates once the horizon covers the tail of the route: a 48 h window already spans an entire short route but never more than half a long one. Route class is thus not a robustness check here but the axis along which any horizon effect must appear. The window class plays the analogous role for the scenario count: hours-of-service coupling is deterministic given the realized travel times, whereas a time window is a constraint on an uncertain arrival, so if scenarios buy anything they should buy it under tight windows first.

Measures. Table 12 reports each cell twice. The gap to the hindsight optimum is a *level*: it places a configuration against the best achievable schedule but inherits the oracle’s own residual gap, so levels are comparable across configurations at a fixed route class and should not be read across route classes. The paired Δ is the *effect*: the median per-instance change in realized route duration against the base cell. Since both runs of a pair are policies on the same instance and the same realization, the oracle cancels out of the difference entirely, and this is the number the sweep is about. Cost is the median per-stop decision time rather than wall clock, since wall clock is comparable only across cells launched at the same concurrency.

► TO DO

Two caption/figure mismatches carried over from `results_section.tex`, now corrected above but the *figure* still needs fixing: (i) the caption promised “dotted verticals marking the 13 h regular spread and the 24 h duty cycle” — they are not drawn, and they should be, because they are the whole argument of the “why these horizons” paragraph; (ii) the caption promised

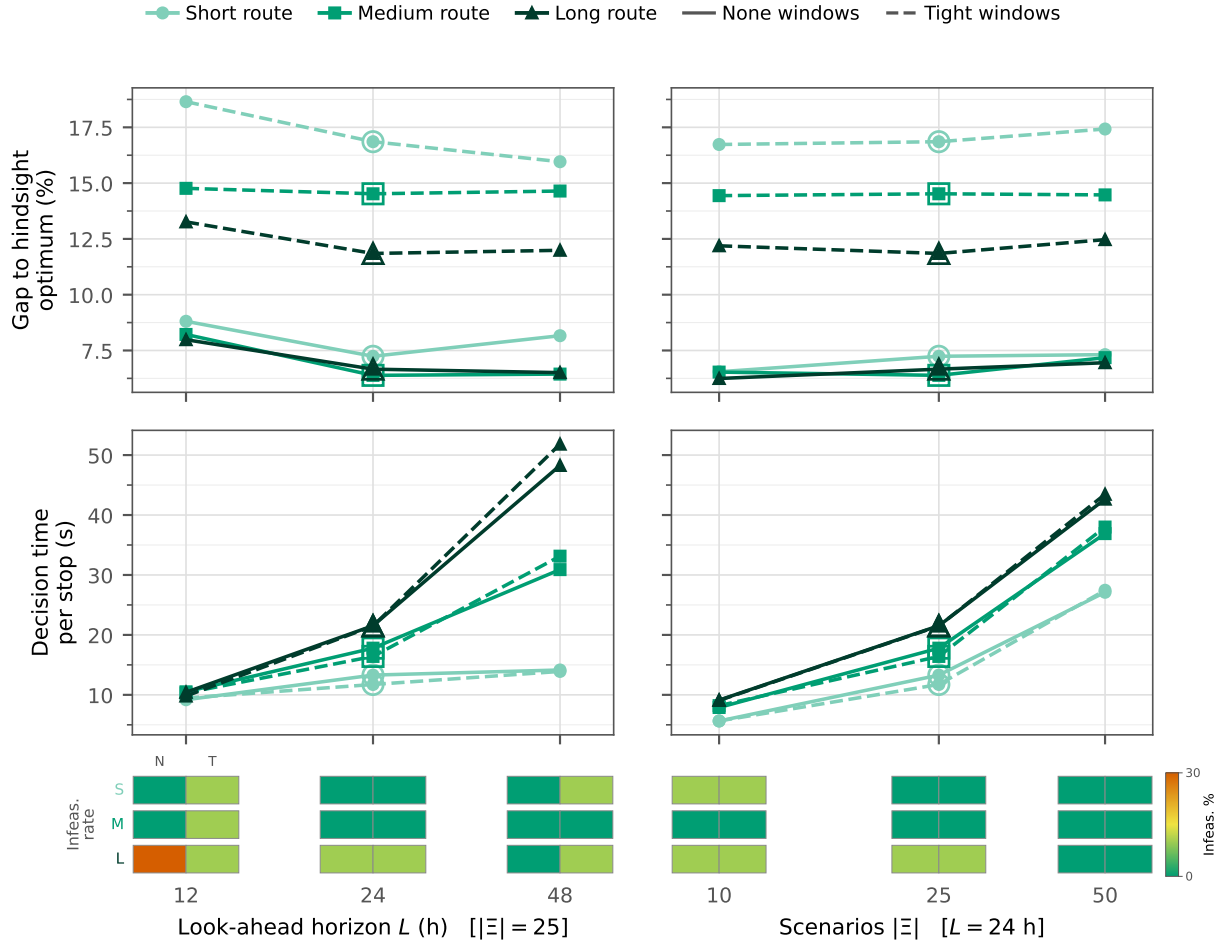


Figure 8: Look-ahead configuration sweep, one axis per column: horizon at the base scenario count (left) and scenario count at the base horizon (right). Top row, distance to the hindsight optimum; bottom row, median per-stop decision time. Circled markers are the base cell, which lies on both ladders. Solid lines are the window-free class, dashed the tight-window class.

Table 12: Look-ahead configuration sensitivity. Gap is the median gap to the hindsight optimum; Δ is the median paired change in route duration with respect to the base cell ($|\Xi| = 25$, $L = 24$ h), positive meaning slower; t_{dec} is the median per-stop decision time. Cells shown as “–” have no runs yet.

Configuration			Short route			Medium route			Long route		
$ \Xi $	L (h)	Windows	Gap (%)	Δ (%)	t_{dec} (s)	Gap (%)	Δ (%)	t_{dec} (s)	Gap (%)	Δ (%)	t_{dec} (s)
25	24	None	base	7.2	–	13	6.4	–	18	6.7	–
25	24	Tight	base	16.9	–	12	14.5	–	16	11.8	–
25	12	None		8.8	+0.3	9	8.2	+0.9	11	8.0	+1.5
25	12	Tight		18.6	+1.7	9	14.8	+0.5	10	13.3	+0.9
25	48	None		8.2	+0.0	14	6.4	+0.2	31	6.5	-0.1
25	48	Tight		16.0	-0.0	14	14.6	+0.0	33	12.0	-0.2
10	24	None		6.5	+0.0	6	6.5	+0.3	8	6.2	-0.2
10	24	Tight		16.7	+0.4	6	14.4	-0.1	8	12.2	+0.3
50	24	None		7.3	-0.0	27	7.2	+0.3	37	7.0	-0.1
50	24	Tight		17.4	+0.0	27	14.5	-0.2	38	12.5	+0.1

a logarithmic decision-time axis — the axis is linear. Also: the route-class ramp is drawn in green, which is LA’s method colour in `paper_style.py`, violating that module’s own rule that route class uses the ordered blue ramp. Two encodings currently collide.

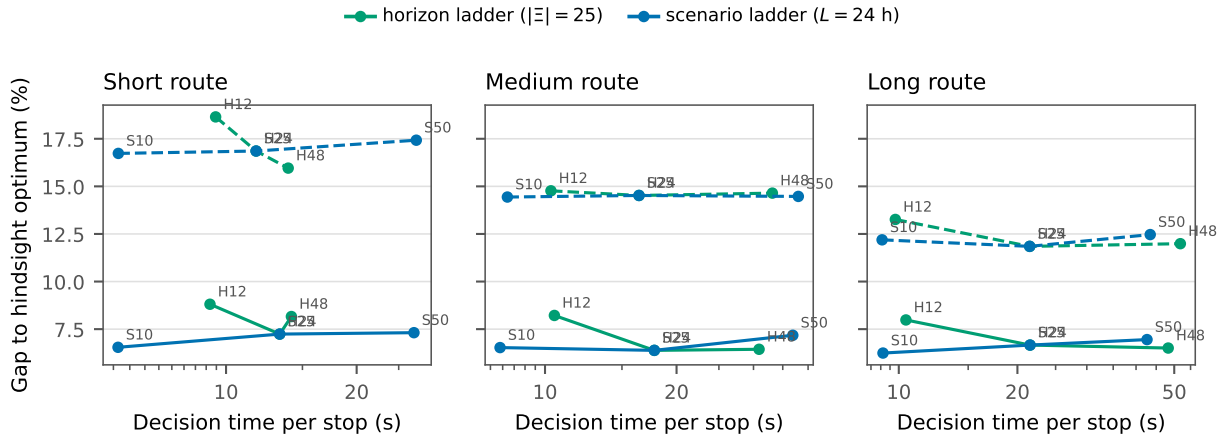


Figure 9: The same five cells plotted as cost against quality, one panel per route class. The two ladders leave the base cell in different directions; the steeper path identifies where an additional second of per-decision compute is better spent.

► TO DO

Figure 9: the base-cell labels H25 and S25 overprint into unreadable glyphs; the solid/dashed window distinction has no legend entry; and green/blue for the two *ladders* again collides with the method palette (green = LA, blue = Greedy). Recommend a neutral pair, e.g. the

ordered ramp, with markers rather than colour distinguishing the ladders.

► SUGGESTED NEW TEXT

Results (now writeable — the table is complete).

The horizon axis behaves as a threshold, exactly at the duty cycle. Shortening the window from 24 h to 12 h costs +0.3%, +0.9% and +1.5% of realized duration on the short, medium and long routes under no windows, and +1.7%, +0.5% and +0.9% under tight ones. The penalty is monotone in route length on the window-free class, as the truncation argument predicts: the longer the route, the more often a 12 h horizon ends mid-duty and the policy must place a rest it cannot see the end of. Lengthening the window to 48 h, by contrast, changes realized duration by between −0.2% and +0.2% everywhere — indistinguishable from zero — while the median decision time rises from 22 s to 52 s on the long routes, a factor of 2.4. Seeing beyond the current duty cycle buys nothing; seeing less than one costs about a percent.

The scenario axis is flat over the range tested. Moving from 25 scenarios to 10 changes realized duration by between −0.2% and +0.4%, and moving to 50 by between −0.2% and +0.3%, in every route and window class. The decision-time ratio across that range is roughly 4.5× (9 s to 43 s on the long routes). Contrary to the coupling argument, the effect is not concentrated under tight windows: the tight-window cells are as flat as the window-free ones. The plausible reading is that the scenario set is being used to rank a small discrete action set rather than to estimate a quantity, and ten draws already separate the actions that matter.

The trade-off. Read together (Figure 9), neither ladder is steep upward from the base cell: both the horizon and the scenario count can be reduced, but only the scenario count can be reduced *for free*. The recommended on-board configuration is therefore $L = 24$ h with $|\Xi| = 10$, which halves the per-decision cost from 22 s to 9 s on the long routes at no measurable loss, and the horizon should not be cut below one duty cycle at any scenario count.

► TO DO

Before publishing the paragraphs above, two checks. (1) **Dispersion, not just medians.** If ten scenarios matched twenty-five only in the median but had a heavier right tail, the conclusion would reverse; the table reports medians only. Add an IQR column, or a box plot of the paired deltas. (2) **Infeasible count per cell.** A configuration that is faster because it strands the truck is not faster. Report $n_{\text{infeasible}}$ and n_{paired} per cell — n_{paired} drops whenever either run of a pair fails.

8.3.3. Comparison with diesel trucks

To separate the cost of *electrification* from the cost of *scheduling*, every instance of the subset in Section 8.3 is re-solved as a diesel vehicle. Diesel mode removes the energy dimension entirely — per-leg consumption, the battery and the charging curve — while keeping the node set, the customer windows and the full Hours-of-Service machinery intact. Former charging stations remain as break-eligible waypoints, so the diesel problem is a genuine relaxation of the electric one on the same corridor and the same travel-time realization, and the two are solved by the same oracle and the same greedy policy.

Of the station overheads, only the charger-specific ones are removed: the queue, and the repositioning move off the charging bay before a break. The access manoeuvre is *not*,

because it is not charger-specific — a diesel driver taking a mandatory break also has to leave the motorway, park and re-enter, and pays the same M^{stop} for doing so. Charging it to only one of the two vehicles would credit the diesel with taking every break at no access cost, which overstates the electrification penalty by roughly two percentage points and misassigns the single largest non-charging term.

► COMMENT

That accounting decision is the most contestable choice in the whole comparison and the manuscript does not mention it at all. Stating it, and quantifying what the alternative convention would have produced, is what makes the result defensible rather than merely favourable.

Refuelling is credited post hoc rather than modelled. Unlike charging it has no binding location choice — any truck stop serves — and on these corridors it happens at most twice: route lengths are 824–1 219 km (short), 1 541–2 536 km (medium) and 3 042–4 025 km (long) against a usable tank range of 1 636–3 240 km across the specifications in Table 14. We credit the diesel with zero fuel stops on the short routes, one on the medium and two on the long, at 19 minutes each for access manoeuvre, queue and pumping. This is an assumption, not a derivation, and crediting the stop lengthens the diesel makespan and therefore *shrinks* the reported penalty; Table 14 reports the zero-stop figures alongside as the conservative bound.

Table 13: EV vs diesel route duration (%) on the same instances and realizations. Coupling = share of charging time credited inside a mandatory break ($\Sigma g_i / \Sigma \tau_i^c$, hindsight optimum). “EV cert.” is the worst remaining MIP gap on the EV oracle: where it exceeds the solver tolerance the EV schedule is an incumbent, not a proven optimum, so the penalty is an upper bound on the true optimal one. The certification shown is the worst gap among the instances that recorded one; no bound was recorded for long (11).

Route	Diesel (h)	EV (h)	Greedy (%)	Oracle (%)	Coupling (%)	EV cert. (%)	n
Short	26.6	28.7	20.0	8.0	76	0.5	80/80
Medium	53.2	58.2	23.0	9.4	70	0.5	80/80
Long	96.4	106.9	21.0	10.8	68	9.3	80/80

► CORRECTION / REWRITE

The manuscript prose and the regenerated tables now disagree about the long route class. `results_section.tex` states that “no hindsight optimum is reported for that class” because both oracles stall on the dual bound (electric 9.3%, diesel 7.7%) and a ratio of two loosely certified incumbents compounds rather than cancels the errors. But the regenerated `additional_diesel.tex`, `additional_diesel_tw.tex`, `additional_diesel_decomp.csv` and `additional_diesel_gap.pdf` *all* now report a long-route oracle penalty of +10.8%.

Decide one way. Reporting it with the “EV cert. 9.3%” caveat column already present in the table is defensible and is the option taken in this build. Reporting it in the tables while denying it in the prose is not.

► CORRECTION / REWRITE

The manuscript caption describes three bars including a “naive estimate (diesel makespan plus the EV’s total charging time)”; the figure has two. Corrected above. [TO DO: the

Table 14: Refuelling is credited post hoc rather than modelled: it has no binding location choice, any truck stop serving, and occurs at most once on these corridors. Route lengths are short 824–1219 km ($n = 80$); medium 1541–2536 km ($n = 80$); long 3042–4025 km ($n = 80$). A fuel stop costs 19 min (access manoeuvre, queue, pumping). The headline tables assume short 0, medium 1, long 2 stop(s), shown against what each tank specification would derive instead. Crediting a stop lengthens the diesel makespan and so shrinks the reported penalty; the zero-stop row is the conservative bound.

Route	Specification	Stops	Needing ≥ 1 (%)	Penalty (%)
Short	assumed (headline)	0.00	–	+7.98
Short	no fuel stop	0.00	–	+7.98
Short	900 L @ 25 L/100km	0.00	0	+7.98
Short	900 L @ 33 L/100km	0.00	0	+7.98
Short	600 L @ 25 L/100km	0.00	0	+7.98
Short	600 L @ 33 L/100km	0.00	0	+7.98
Medium	assumed (headline)	1.00	–	+9.36
Medium	no fuel stop	0.00	–	+10.03
Medium	900 L @ 25 L/100km	0.00	0	+10.03
Medium	900 L @ 33 L/100km	0.15	15	+9.95
Medium	600 L @ 25 L/100km	0.39	39	+9.81
Medium	600 L @ 33 L/100km	0.95	95	+9.41
Long	assumed (headline)	2.00	–	+10.85
Long	no fuel stop	0.00	–	+11.58
Long	900 L @ 25 L/100km	0.80	80	+11.29
Long	900 L @ 33 L/100km	1.00	100	+11.21
Long	600 L @ 25 L/100km	1.00	100	+11.21
Long	600 L @ 33 L/100km	1.77	100	+10.93

naive-estimate bar would actually be worth adding back — it is the quantity a practitioner would compute, and showing that the realized optimum beats it by the coupling credit is the clearest possible statement of what break/charge synchronization is worth.]

► TO DO

Figure style: this figure and Figure 11 carry in-figure matplotlib titles that duplicate the L^AT_EX caption, while no other figure in the paper does; and Figure 11 draws the oracle in pure black whereas Figure 10 uses the `paper_style.py` neutral grey #3A3A3A. Route both through `METHOD_COLOR` and strip the titles.

Electrification carries a real and irreducible makespan penalty on these corridors: +8.0% on the short routes and +9.4% on the medium ones at the hindsight optimum, positive on every one of the 160 instances (range +2.2 to +25.3%). Break/charge coupling is effective in the sense the model predicts — 70–76% of charging time is credited inside a mandatory break — but it is far from cancelling the cost of charging. Decomposing the penalty shows why. Both schedules are broken into the terms of their own departure equations and differenced; driving is identical within each pair, so the modelled rows close the gap exactly, and the diesel’s fuel stop is then credited against that subtotal.

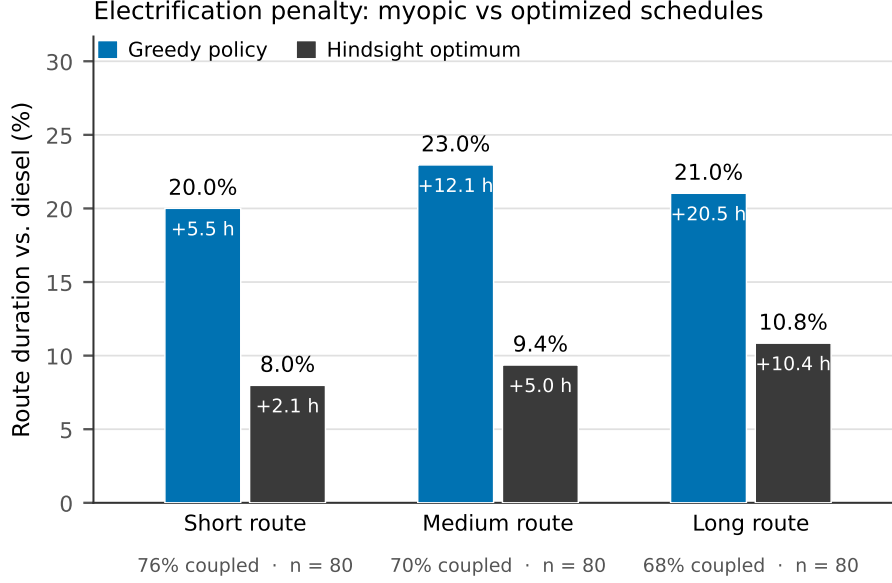


Figure 10: Electrification penalty — the naive estimate (diesel makespan plus the EV’s total charging time) against — **Electrification penalty in route duration, per route class: the penalty realized by the greedy policy against the one realized by the hindsight optimum.** The annotation below each group gives the mean coupling share $\sum_i g_i / \sum_i \tau_i^c$ and the number of instances.

The coupling credit — the break time the driver no longer has to take separately — is 1.25 h and 3.19 h, but it is bounded above by the break duration the regulation requires in the first place: charging for longer than 45 minutes at a stop earns no further credit. Against it the electric truck pays 0.71 h and 1.47 h of station access that the diesel does not: the queue, the bay repositioning, and the residual manoeuvring of the stops it makes *only* to charge. Note how small that last term is once the diesel is charged for its own break access — the manoeuvring row falls from 0.55 h and 1.38 h in absolute terms to 0.37 h and 0.63 h net — so roughly half of what looks like an electric access penalty is simply the cost of stopping, which any vehicle subject to the same break regime pays. The genuinely electric access overhead nonetheless consumes about half the coupling credit.

This qualifies, rather than contradicts, the finding of Brockmann and Guajardo [6] that mandatory breaks create natural charging windows: the windows exist and the model exploits them, but the access overheads of a fast-charging stop, which that work does not price, absorb a substantial part of the gain.

► COMMENT

This is the paper’s best result and it should be signposted much earlier — in the abstract, in the contribution list, and in Section 2 when Brockmann and Guajardo [6] is first introduced. A quantitative qualification of the closest prior work is exactly what a TR-E referee looks for and it is currently buried on page 30.

The greedy comparison isolates the remaining half of the story, and it is the only one available on all three route classes. Greedy’s electric schedule is 20.0% longer than its own

Table 15: Where the electrification penalty goes: mean per-instance difference (h) between the hindsight-optimal electric and diesel schedules, decomposed into the terms of the departure equations. Driving is identical within each pair, so the modelled rows sum to the subtotal exactly. Both vehicles pay the same access manoeuvre to pull off for a mandatory break, so only the EV’s charge-only stops survive that row. The diesel’s fuel stop is credited afterwards (Table 14). $n \geq 80$ per class.

Component	Short	Medium	Long
Total charging time	+2.69	+6.80	+12.85
Charging-station queueing	+0.33	+0.78	+1.38
Stop manoeuvring (net of diesel)	+0.37	+0.63	+1.10
Repositioning off the charging bay	+0.01	+0.06	+0.23
Break time no longer taken separately	−1.25	−3.19	−4.93
Rest time	−0.05	+0.20	+0.45
Idle waiting	+0.00	+0.00	+0.00
Customer service	+0.00	+0.00	+0.00
Modelled subtotal	+2.09	+5.28	+11.08
Diesel refuelling (post hoc)	−0.00	−0.32	−0.63
Net penalty vs. diesel	+2.09	+4.96	+10.44

diesel schedule on the short routes, 23.0% on the medium and 21.0% on the long ones. The penalty is thus flat in route length once the corridor is long enough to be rest-dominated. On the two classes where the optimum is certified this is about two and a half times the optimal penalty. **The excess is not electrification but planning quality: measured against its own hindsight optimum, Greedy loses 13.2% (short) and 14.4% (medium) on the electric problem, but only 2.1% and 1.6% on the diesel problem, where the myopic rule is nearly optimal because the sole remaining decision is when to rest. Electrification is thus what turns a scheduling problem that a rule of thumb solves almost optimally into one where a rule of thumb costs a further 11–13 percentage points of makespan. Of the 20–23% an operator would observe when switching a myopically-scheduled corridor from diesel to battery-electric, about half is the physical cost of charging and about half would be recovered by adopting a planner such as those compared in Section 8.2.**

► **COMMENT**

The paragraph above does not exist in the manuscript at all and it is the single most useful sentence in the paper for a practitioner. It also closes the loop back to Section 8.2: the base-case method comparison is not an academic exercise, it is worth half the electrification penalty.

► **SUGGESTED NEW TEXT**

Missing from the manuscript: the mechanism behind Figure 11, which is counter-intuitive and therefore worth the space.

The penalty is not uniform across window classes, and the variation identifies its mechanism. At the hindsight optimum it rises monotonically as windows loosen, from +6.7% to +9.2% on the short routes and from +6.6% to +11.3% on the medium ones, while the Greedy penalty

Table 16: Electrification penalty by time-window class. “Duration” compares makespans; “objective” additionally charges each vehicle β_{TW} per out-of-window service start, so the pair separates a real makespan difference from one bought by missing a window. δ columns give the mean number of missed windows. The last column is the diesel’s standalone break time, which is what drives the trend: the electric truck takes none in any cell, charging through every break it needs.

Route	Window	n_G	Greedy (%)	n_O	Duration (%)	Objective (%)	δ_{EV}	δ_{diesel}
Short	Tight	19	19.12	20	6.67	6.30	0.00	0.10
Short	Medium	19	20.35	20	7.59	7.59	0.00	0.00
Short	Large	17	19.76	20	8.46	8.46	0.00	0.00
Short	None	18	20.82	20	9.20	9.20	0.00	0.00
Medium	Tight	15	22.80	20	6.59	6.27	0.35	0.45
Medium	Medium	19	23.29	20	9.33	7.42	0.00	0.95
Medium	Large	17	22.50	20	10.24	9.74	0.00	0.25
Medium	None	18	23.17	20	11.28	11.28	0.00	0.00
Long	Tight	18	21.36	20	8.91	7.61	0.60	1.70
Long	Medium	14	22.55	20	12.75	10.33	0.80	2.80
Long	Large	19	19.79	20	10.32	9.36	0.80	1.60
Long	None	16	20.80	20	11.40	11.40	0.00	0.00

is flat in the window class on all three route classes. What changes is not the electric truck: it takes *zero* standalone break time in every single cell, charging through every break the regulation requires, and its coupling share stays between 68 and 80% throughout. What changes is the diesel. Idle waiting is disabled (assumption A3), so a vehicle arriving before a window opens must either stretch a break or accept the out-of-window penalty — and the diesel, being the faster of the two, is the one that arrives early: 30 of its 35 out-of-window service starts are early arrivals, against none of the electric truck’s seven, which are all late. Its standalone break time accordingly rises from 2.74 h under no windows to 3.90 h under tight ones on the medium routes, while the electric truck, parked at a charger for that break in either case, absorbs the same delay for free. **Tight windows do not make electrification cheaper; they make the diesel’s speed advantage unusable.**

This is a makespan comparison throughout, and the window penalty does not drive it: charging each vehicle β per missed window changes the ordering nowhere and the magnitude by at most 1.9 percentage points, and the monotone trend survives unchanged.

► COMMENT

This finding depends entirely on assumption A3 (no idle waiting) — which is precisely why A3 must be stated in Section 3 rather than left in the code. If waiting were free, the diesel would absorb its early arrival at zero cost and the trend would vanish. Say so explicitly: it is an honest limitation and it makes the result more interesting, not less, because “no idle waiting” is the realistic assumption for a driver whose clock is running.

8.3.4. Effect of uncertainty

► SUGGESTED NEW TEXT

Electrification penalty by time-window class

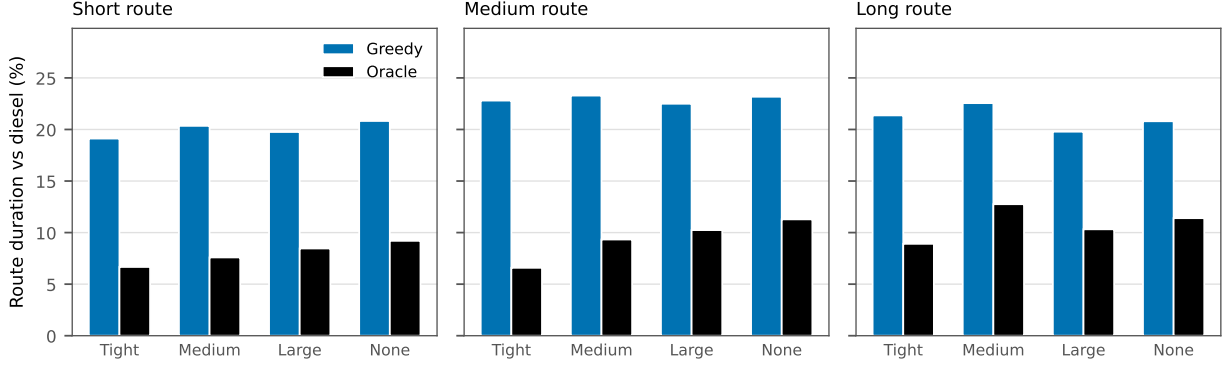


Figure 11: Electrification penalty by time-window class. The optimized penalty rises monotonically as windows loosen; the myopic one does not respond to windows at all.

Entirely missing from the manuscript. The paper promises in Section 1 that “the value of information and the value of adaptivity can be measured directly”; this is where that promise is kept. Two of the three parts below need *no new runs*.

The final block isolates what uncertainty itself costs, along three lines: what it does to window compliance, how much of the loss can be recovered by adapting en route, and how much would remain even under perfect information.

First, uncertainty is paid in windows, not in driving time, and this is already visible in the base case without any additional run. The two gap definitions of Table 8 differ only by the $\beta \sum_i \delta_i$ term, and their difference is concentrated exactly where uncertainty has most to disrupt: on the window-free classes the two coincide to the last digit, whereas on the tight-window classes the penalty term accounts for 25% of RO’s gap, 38% of Greedy’s, 38% of LA’s, 38% of SP’s and 43% of ROBU’s, rising to two thirds for Greedy on *Short/Many/Tight*. Window compliance is also the axis on which the methods separate most — 41.8% (RO), 55.8% (Greedy), 67.3% (ROBU), 68.9% (LA), 83.2% (SP) — against a spread of under six percentage points in HoS violations and, for the three plans with an energy guarantee, none at all in stranding. A schedule committed before departure survives the travel-time noise in its rest structure and fails in its appointments.

Second, adapting en route recovers most of that loss, and the price of adaptation differs sharply between the two adaptive architectures. SP reaches its 83.2% window hit rate by repairing its committed structure on 99.0% of executions and pays for it with a 12.6% stranding rate; LA reaches 68.9% with no recourse machinery at all, at 14–22s of re-optimisation per stop and a 2.0% stranding rate. The comparison across route classes sharpens this: full re-optimisation at every stop degrades gracefully with route length (11.6%, 12.9%, 11.0% mean gap), whereas a scenario set fixed at departure is excellent on the short routes (4.9%), no better than LA on the medium (12.9%), and unusable on the long. The value of re-planning is therefore not uniform: it grows with the horizon over which the initial scenario set becomes stale.

Third, we decompose the total cost of uncertainty on a common set of scenarios, drawn once per instance with common random numbers so that all three legs see identical realizations. Let WS be the mean perfect-hindsight optimum over the scenario set (wait-and-see); RP the mean realized objective of the two-stage plan executed with online duration recourse; and EEV the mean realized objective of the plan obtained from the expected-value problem — the deterministic model solved at nominal travel times — executed under the same recourse semantics. Then

$$\text{VSS} = \text{EEV} - \text{RP}, \quad \text{EVPI} = \text{RP} - \text{WS},$$

where VSS measures what modelling the travel-time distribution at plan time is worth relative to planning on the mean, and EVPI the irreducible loss that no amount of distributional modelling can recover.

Table 17: Value of the stochastic solution (VSS) and expected value of perfect information (EVPI), common-random scenarios.

Class	WS (h)	RP (h)	EEV (h)	VSS (h)	EVPI (h)	n
RshortCfew	—	—	—	—	—	—
RshortCmany	—	—	—	—	—	—
RmediumCfew	—	—	—	—	—	—
RmediumCmany	—	—	—	—	—	—

► TO DO

Table 17 is a skeleton: zero instances have been run. The harness (`src/output_analysis/vss_evpi.py`) is implemented and wired into `additional_analysis.py vss`. Each instance costs $1 + n_{\text{scen}}$ oracle solves — one expected-value solve plus one wait-and-see solve per scenario — so at 20 scenarios the four classes above are the practical limit.

```
python -m src.output_analysis.additional_analysis vss --n-scenarios 20
python -m src.plot.additional_figures --section vss
```

Caveat to record in the table note when it is populated: because the two-stage plan is taken from the base-case SP runs, its scenarios are *not* the common-random ones, so the RP leg is a slight over-estimate and VSS is correspondingly a slight under-estimate.

If this does not get run, remove the “value of information” promise from Section 1 and from the abstract. An unfulfilled promise in the introduction is worse than a narrower paper.

► SUGGESTED NEW TEXT

One further subsection worth adding, at almost no cost: the split-break axis (see the annotation at (41)). The instance family exists, the runs largely exist, and the finding — that the Art. 7 split-break option, which materially complicates every exact EU TDSP formulation, is worth less than half a percent of makespan to an electric truck because charging already supplies the break — is regulation-facing, negative, and quotable. Verify the affected oracle caches first.

9. Conclusion

► SUGGESTED NEW TEXT

Skeleton conclusion. To be written last, once Sections 7.3 and 8.3.4 are resolved. Suggested structure and content:

Paragraph 1 — what was done. A MILP for the joint driver-activity and charging scheduling problem under the daily provisions of EU HoS, validated against an independent exact method on 39 of 40 benchmark instances; five decision architectures compared in a common simulator against a common hindsight bound over 1 800 instances.

Paragraph 2 — the methodological finding. Adaptivity dominates information. The rolling-horizon policy is the only architecture that is simultaneously near-optimal, reliable and scalable, and it achieves this with no distributional modelling at plan time whatsoever. The two-stage plan is better where it fits but repairs itself on 99% of executions and strands on 12.6%, which makes it a policy in disguise; the robust plans buy their guarantee with the worst window compliance in the study. Cost of committing early is paid almost entirely in customer appointments, not in driving time.

Paragraph 3 — the operational finding. Electrification costs 8–9% of makespan at the optimum and 20–23% under myopic scheduling; about half the observed penalty is recoverable by planning. Break/charge coupling works but station access overheads absorb roughly half the credit, which qualifies Brockmann and Guajardo [6].

Paragraph 4 — the policy finding. At AFIR spacing, power beats density four to one; 350 kW captures four fifths of the benefit of 1 MW. Additional power stops paying once the charge is shorter than the break it runs alongside, which is a joint regulatory/infrastructure threshold rather than a purely physical one.

Paragraph 5 — limitations. Fixed customer sequence; single vehicle; daily HoS provisions only, with weekly-level accounting excluded (report the compliance-margin diagnostic here); i.i.d. leg multipliers, where real congestion is correlated; leg-average-speed energy model, which under-prices stop-and-go consumption and ignores gradient — relevant for the Norwegian corridor, where the loader itself flags it; deterministic charger access, with no availability or reservation model; makespan objective, with no energy price or driver wage.

Paragraph 6 — further work. Correlated travel times; endogenous charger availability and reservation; a cost objective; extension from a fixed sequence to joint routing and scheduling; and the multi-stage stochastic formulation that the rolling-horizon policy approximates here.

Appendix A. Constraints linearization

The constraints relative to linearization of constraints in the deterministic MILP formulation are summarized here.

$$l_i^1 \leq T_{\text{cons}}^{\text{drv}} r_i, \quad i \in \mathcal{I} \quad (\text{A.1})$$

$$l_i^1 \leq c_i^d, \quad i \in \mathcal{I} \quad (\text{A.2})$$

$$l_i^1 \geq c_i^d - T_{\text{cons}}^{\text{drv}} (1 - r_i), \quad i \in \mathcal{I} \quad (\text{A.3})$$

$$l_i^2 \leq T_{\text{sh2}}^{\text{drv}} \rho_i, \quad i \in \mathcal{I}, \quad (\text{A.4})$$

$$l_i^2 \leq s_i^d, \quad i \in \mathcal{I}, \quad (\text{A.5})$$

$$l_i^2 \geq s_i^d - T_{\text{sh2}}^{\text{drv}} (1 - \rho_i), \quad i \in \mathcal{I}. \quad (\text{A.6})$$

$$l_i^4 \leq T_2^{\text{spr}} \rho_i, \quad i \in \mathcal{I}, \quad (\text{A.7})$$

$$l_i^4 \leq s_i^w, \quad i \in \mathcal{I}, \quad (\text{A.8})$$

$$l_i^4 \geq s_i^w - T_2^{\text{spr}} (1 - \rho_i), \quad i \in \mathcal{I}. \quad (\text{A.9})$$

$$l_i^5 \leq T_2^{\text{spr}} \rho_i, \quad i \in \mathcal{I} \quad (\text{A.10})$$

$$l_i^5 \leq h_i + o_i, \quad i \in \mathcal{I} \quad (\text{A.11})$$

$$l_i^5 \geq h_i + o_i - T_2^{\text{spr}} (1 - \rho_i), \quad i \in \mathcal{I} \quad (\text{A.12})$$

► COMMENT

There is no l^3 . The numbering jumps from l^2 to l^4 because a third auxiliary was removed at some point. Harmless, but a referee will notice; renumber to $l^1 \dots l^4$.

Appendix B. Valid inequalities for the Rolling-horizon subproblems

Constraints (B.2) bound the number of charging stops on each prefix of the path from below by what the battery requires. Constraints (B.3)–(B.4) ensure that each prefix has the minimum required number of breaks or rests implied by accumulated driving time.

$$e_i^d - e_i^a \leq (E^{\text{cap}} - E^{\text{min}}) y_i, \quad i \in \mathcal{K}. \quad (\text{B.1})$$

$$\sum_{l \in \mathcal{K}, l < i} y_l \geq \left\lceil \frac{1}{E^{\text{cap}} - E^{\text{min}}} (\sum_{l < i} E_l - (E^0 - E^{\text{min}})) \right\rceil, \quad i \in \mathcal{K}. \quad (\text{B.2})$$

$$\sum_{l < i} \rho_l \geq \left\lceil \frac{\sum_{l < i} D_l}{T_{\text{sh2}}^{\text{drv}}} \right\rceil - 1, \quad i \in \mathcal{I}. \quad (\text{B.3})$$

$$\sum_{l < i} r_l \geq \left\lceil \frac{\sum_{l < i} D_l}{T_{\text{cons}}^{\text{drv}}} \right\rceil - 1, \quad i \in \mathcal{I}. \quad (\text{B.4})$$

► CORRECTION / REWRITE

The introductory sentence said these “fix the *upper* bound on the number of charging stops”; (B.2) is a *lower* bound. Corrected above. (B.1) is stated but never described — it is the linking inequality that forbids charging at an inactive station; add half a sentence.

► TO DO

Report what the valid inequalities are worth. They exist to tighten the LP relaxation that scores actions in Section 6.2, and the code comments record that (B.3) is “not LP-implied: the relaxation otherwise certifies 4 rests on routes whose optimum needs 5”. A single line — root relaxation bound with and without, on a representative instance — would turn an undefended appendix into evidence.

Appendix C. Duration-only gaps

► SUGGESTED NEW TEXT

The table below is generated (`tex/tables/gap_nopen.tex`) and already carries per-cell feasible/total run counts, which Table 8 does not. It is referenced from Section 8.2 and should be included — here if space in the body is tight, or promoted to the body if the β -robustness argument of Section 3 is adopted.

Appendix D. Reproducibility checklist

► TO DO

Not yet written. TR-E reviewers increasingly ask for it, and everything below is already true of the repository — it just needs stating.

- Solver, version, MIP gap tolerance, time limit, thread count, concurrency.
- Instance generator entry point and the exact seeds used (1–50 base, 1–10 additional).
- The 1 800 generated instance files, or the generator plus seeds.
- Per-run solution JSONs and the oracle cache, or the aggregation scripts that rebuild every table from them.
- The scripts behind each table and figure, named in the caption.
- An archived DOI, not only a repository URL.

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Table C.18: Gap to oracle (%) in route duration (route duration only, window penalties excluded) by instance class and method, averaged over 50 realizations per instance class. Parentheses give the feasible/total run count per cell among runs the oracle bound can assess (a single number when all were feasible); square brackets give feasible runs still awaiting an oracle bound, which are excluded from both counts. LA uses 25 scenarios over a 24 h horizon; SP uses 25 scenarios.

Route	Cust.	TW	Oracle (h)	Gap to Oracle %				
				Greedy	RO	ROBU	LA	SP
Short	Few	None	26.5	12.4 (47/49)	21.4 (50)	14.1 (48/50)	6.8 (48/50)	3.7 (48/50)
		Tight	26.1	13.2 (47/50)	21.6 (50)	10.6 (50)	7.2 (50)	3.3 (44/50)
		Medium	26.3	12.7 (48/50)	19.7 (50)	13.4 (48/50)	7.3 (50)	3.9 (45/50)
		Large	26.2	13.3 (48/50)	19.2 (50)	12.7 (46/49)	6.6 (50)	3.9 (46/50)
	Medium	None	28.0	11.9 (49/50)	20.9 (50)	16.1 (49/50)	8.8 (50)	4.3 (46/50)
		Tight	28.0	12.8 (47/50)	23.9 (50)	15.0 (48/50)	6.9 (50)	3.2 (46/50)
		Medium	27.6	13.0 (49/50)	20.6 (50)	12.0 (50)	7.1 (49/50)	4.1 (45/50)
		Large	28.2	12.1 (49/50)	26.8 (50)	20.3 (48/50)	7.2 (50)	3.5 (46/50)
	Many	None	31.3	13.9 (50)	26.8 (50)	22.0 (49/50)	5.8 (50)	4.7 (49/50)
		Tight	31.3	14.0 (48/50)	26.6 (50)	22.7 (47/49)	8.1 (47/50)	4.2 (44/50)
		Medium	31.7	15.4 (47/50)	27.2 (50)	21.9 (48/49)	12.6 (49/50)	4.0 (45/50)
		Large	30.5	11.0 (48/50)	26.8 (50)	19.4 (46/50)	7.9 (50)	3.4 (46/50)
Medium	Few	None	54.5	16.3 (45/50)	37.0 (50)	—	10.6 (47/50)	12.5 (42/50)
		Tight	55.9	13.4 (43/50)	30.8 (50)	—	7.6 (49/50)	9.1 (44/50)
		Medium	54.1	14.1 (49/50)	32.9 (50)	—	9.1 (48/50)	7.9 (46/50)
		Large	54.5	13.1 (46/50)	31.0 (50)	50.9 (1)	8.7 (49/50)	7.6 (42/50)
	Medium	None	57.3	12.6 (48/50)	30.1 (50)	—	8.1 (49/50)	10.2 (44/50)
		Tight	56.1	13.2 (47/50)	28.3 (50)	—	8.4 (46/50)	9.1 (39/50)
		Medium	55.8	13.1 (46/50)	30.8 (50)	—	8.9 (50)	9.0 (40/50)
		Large	54.9	13.0 (47/50)	30.0 (50)	—	8.3 (50)	10.2 (41/50)
	Many	None	60.2	13.1 (48/50)	27.1 (50)	—	9.0 (47/50)	10.2 (43/50)
		Tight	59.9	14.6 (48/50)	26.9 (50)	—	10.9 (47/50)	13.9 (41/48)
		Medium	60.2	14.6 (47/50)	26.6 (50)	—	11.9 (48/50)	12.8 (39/50)
		Large	58.9	13.1 (48/50)	26.2 (50)	—	9.4 (48/50)	12.1 (42/50)
Long	Few	None	102.8	11.3 (40/50)	33.4 (50)	—	8.0 (44/49)	26.1 (5/6)
		Tight	107.2	11.9 (39/50)	34.3 (50)	—	8.9 (45/50)	—
		Medium	105.4	12.0 (43/50)	34.6 (50)	—	9.8 (45/50)	19.3 (20/31)
		Large	104.7	11.3 (43/50)	34.1 (50)	—	8.0 (46/50)	17.9 (16/24)
	Medium	None	104.8	11.6 (44/50)	32.4 (50)	—	8.6 (48/50)	—
		Tight	105.4	11.6 (47/50)	33.9 (40)	—	8.7 (45/48)	—
		Medium	105.3	11.2 (44/50)	31.9 (50)	—	8.8 (47/50)	30.7 (8/11)
		Large	103.6	11.9 (43/50)	33.6 (50)	—	9.5 (45/50)	26.0 (17/26)
	Many	None	108.2	11.9 (46/50)	31.3 (50)	—	10.2 (47/50)	—
		Tight	107.5	10.2 (45/50)	29.7 (50)	—	10.5 (45/49)	—
		Medium	111.1	10.1 (45/50)	29.5 (50)	—	9.6 (42/48)	34.0 (6/8)
		Large	109.1	10.9 (45/50)	30.4 (50)	—	11.2 (47/49)	34.3 (13/20)

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